MARKING SCHEME

MATHEMATICS (Subject Code–041)

(PAPER CODE: 65/1/1)

Section A

Q.No.	EXPECTED OUTCOMES/VALUE POINTS	Marks
	SECTION A Questions no. 1 to 18 are multiple choice questions (MCQs) and questions number 19 and 20 are Assertion-Reason based questions of **1 mark each.**	
1.	A function $f : R_+ \to R$ (where R_+ is the set of all non-negative real numbers) defined by $f(x) = 4x + 3$ is : (A) one-one but not onto (B) onto but not one-one (C) both one-one and onto (D) neither one-one nor onto	
Sol.	**(A) one-one but not onto**	1
2.	If a matrix has 36 elements, the number of possible orders it can have, is : (A) 13 (B) 3 (C) 5 (D) 9	
Sol.	**(D) 9**	1
3.	Which of the following statements is true for the function $f(x) = \begin{cases} x^2 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases}$? (A) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R}$ (B) $f(x)$ is continuous $\forall\, x \in \mathbb{R}$ (C) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R} - \{0\}$ (D) $f(x)$ is discontinuous at infinitely many points	
Sol.	**(C) $f(x)$ is continuous and differentiable $\forall\, x \in R - \{0\}$**	1

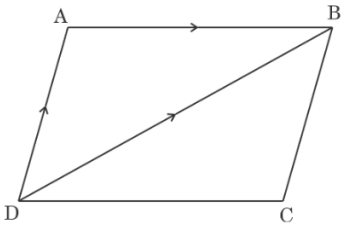
4.	Let f(x) be a continuous function on [a, b] and differentiable on (a, b). Then, this function f(x) is strictly increasing in (a, b) if (A) $f'(x) < 0, \forall\, x \in (a, b)$ (B) $f'(x) > 0, \forall\, x \in (a, b)$ (C) $f'(x) = 0, \forall\, x \in (a, b)$ (D) $f(x) > 0, \forall\, x \in (a, b)$	
Sol.	**(B) $f'(x) > 0, \forall\, x \in (a, b)$**	**1**
5.	If $\begin{bmatrix} x+y & 2 \\ 5 & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$, then the value of $\left(\frac{24}{x} + \frac{24}{y}\right)$ is : (A) 7 (B) 6 (C) 8 (D) 18	
Sol.	**(D) 18**	**1**
6.	$\int_a^b f(x)\,dx$ is equal to : (A) $\int_a^b f(a-x)\,dx$ (B) $\int_a^b f(a+b-x)\,dx$ (C) $\int_a^b f(x-(a+b))\,dx$ (D) $\int_a^b f((a-x)+(b-x))\,dx$	
Sol.	**(B) $\int_a^b f(a+b-x)dx$**	**1**
7.	Let θ be the angle between two unit vectors $\hat{a}$ and $\hat{b}$ such that $\sin\theta = \frac{3}{5}$ Then, $\hat{a} \cdot \hat{b}$ is equal to : (A) $\pm\frac{3}{5}$ (B) $\pm\frac{3}{4}$ (C) $\pm\frac{4}{5}$ (D) $\pm\frac{4}{3}$	
Sol.	**(C) $\pm\frac{4}{5}$**	**1**
8.	The integrating factor of the differential equation $(1-x^2)\frac{dy}{dx} + xy = ax$, $-1 < x < 1$, is : (A) $\frac{1}{x^2-1}$ (B) $\frac{1}{\sqrt{x^2-1}}$ (C) $\frac{1}{1-x^2}$ (D) $\frac{1}{\sqrt{1-x^2}}$	
Sol.	**(D) $\frac{1}{\sqrt{1-x^2}}$**	**1**

9.	If the direction cosines of a line are $\sqrt{3}\,k$, $\sqrt{3}\,k$, $\sqrt{3}\,k$, then the value of k is : (A) ± 1 (B) $\pm\sqrt{3}$ (C) ± 3 (D) $\pm\frac{1}{3}$	
Sol.	(D) $\pm\frac{1}{3}$	1
10.	A linear programming problem deals with the optimization of a/an : (A) logarithmic function (B) linear function (C) quadratic function (D) exponential function	
Sol.	(B) linear function	1
11.	If $P(A \mid B) = P(A' \mid B)$, then which of the following statements is true ? (A) $P(A) = P(A')$ (B) $P(A) = 2\,P(B)$ (C) $P(A \cap B) = \frac{1}{2}\,P(B)$ (D) $P(A \cap B) = 2\,P(B)$	
Sol.	(C) $P(A \cap B) = \frac{1}{2}\,P(B)$	1
12.	$\begin{vmatrix} x+1 & x-1 \\ x^2+x+1 & x^2-x+1 \end{vmatrix}$ is equal to : (A) $2x^3$ (B) 2 (C) 0 (D) $2x^3-2$	
Sol.	(B) 2	1
13.	The derivative of $\sin(x^2)$ w.r.t. x, at $x = \sqrt{\pi}$ is : (A) 1 (B) -1 (C) $-2\sqrt{\pi}$ (D) $2\sqrt{\pi}$	
Sol.	(C) $-2\sqrt{\pi}$	1
14.	The order and degree of the differential equation $\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = \frac{d^2y}{dx^2}$ respectively are : (A) 1, 2 (B) 2, 3 (C) 2, 1 (D) 2, 6	
Sol.	(C) 2, 1	1

15.	The vector with terminal point A (2, – 3, 5) and initial point B (3, – 4, 7) is : (A) $\hat{i} - \hat{j} + 2\hat{k}$ (B) $\hat{i} + \hat{j} + 2\hat{k}$ (C) $-\hat{i} - \hat{j} - 2\hat{k}$ (D) $-\hat{i} + \hat{j} - 2\hat{k}$	
Sol.	(D) $-\hat{\imath} + \hat{\jmath} - 2\hat{k}$	1
16.	The distance of point P(a, b, c) from y-axis is : (A) b (B) b^2 (C) $\sqrt{a^2 + c^2}$ (D) $a^2 + c^2$	
Sol.	(C) $\sqrt{a^2 + c^2}$	1
17.	The number of corner points of the feasible region determined by constraints $x \geq 0$, $y \geq 0$, $x + y \geq 4$ is : (A) 0 (B) 1 (C) 2 (D) 3	
Sol.	(C) 2	1
18.	If A and B are two non-zero square matrices of same order such that $(A + B)^2 = A^2 + B^2$, then : (A) AB = O (B) AB = – BA (C) BA = O (D) AB = BA	
Sol.	(B) AB = – BA	1
	Questions No. **19** & **20**, are Assertion (A) and Reason (R) based questions carrying **1** mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below : (A) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	
19.	*Assertion (A) :* For matrix $A = \begin{bmatrix} 1 & \cos\theta & 1 \\ -\cos\theta & 1 & \cos\theta \\ -1 & -\cos\theta & 1 \end{bmatrix}$, where $\theta \in [0, 2\pi]$, $|A| \in [2, 4]$. *Reason (R) :* $\cos\theta \in [-1, 1], \forall\, \theta \in [0, 2\pi]$.	
Sol.	(A) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of Assertion (A).	1

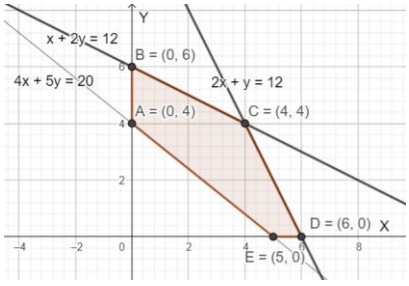
20.	*Assertion (A) :* A line in space cannot be drawn perpendicular to x, y and z axes simultaneously. *Reason (R) :* For any line making angles, α, β, γ with the positive directions of x, y and z axes respectively, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.	
Sol.	(A) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of Assertion (A).	1
	SECTION B In this section there are 5 very short answer type questions of 2 marks each.	
21(a).	Check whether the function $f(x) = x^2\,|x|$ is differentiable at x = 0 or not.	
Sol.	$f(x) = \begin{cases} x^3, x \geq 0 \\ -x^3, x \leq 0 \end{cases}$ $RHD = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} h^2 = 0$ $LHD = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0}(-h^2) = 0$ $\because RHD = LHD = 0$, $So\ f(x)\ is\ differentiable\ at\ x = 0$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
	OR	
21(b).	If $y = \sqrt{\tan\sqrt{x}}$, prove that $\sqrt{x}\ \frac{dy}{dx} = \frac{1+y^4}{4y}$.	
Sol.	$y = \sqrt{tan\sqrt{x}}$ $\frac{dy}{dx} = \frac{sec^2\sqrt{x}}{2\sqrt{tan\sqrt{x}}} \times \frac{1}{2\sqrt{x}}$ $\sqrt{x}\frac{dy}{dx} = \frac{sec^2\sqrt{x}}{4\sqrt{tan\sqrt{x}}}$ $= \frac{1+(tan\sqrt{x})^2}{4\sqrt{tan\sqrt{x}}} = \frac{1+y^4}{4y}$	1 1
22.	Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ has neither maxima nor minima.	
Sol.	$f'(x) = 12x^2 - 36x + 27$ $= 3\,(2x-3)^2 \geq 0$ for all $x \in R$ $\therefore$ f is increasing on R. Hence f(x) does not have maxima or minima.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

23(a).	Find : $\int x\sqrt{1+2x}\ dx$	
Sol.	$1+2x=t^2$ $2\,dx=2t\,dt$ $\frac{1}{2}\int(t^4-t^2)dt=\frac{1}{2}\left[\frac{t^5}{5}-\frac{t^3}{3}\right]+C$ $=\frac{(1+2x)^{\frac{5}{2}}}{10}-\frac{(1+2x)^{\frac{3}{2}}}{6}+C$	$\frac{1}{2}$ 1 $\frac{1}{2}$
	OR	
23(b).	Evaluate : $\int_0^{\frac{\pi^2}{4}}\frac{\sin\sqrt{x}}{\sqrt{x}}\ dx$	
Sol.	$\int_0^{\frac{\pi^2}{4}}\frac{sin\sqrt{x}}{\sqrt{x}}\,dx$ Put $\sqrt{x}=t\ \Rightarrow\ dx=2t\,dt$ $2\int_0^{\frac{\pi}{2}}\sin t\,dt=2\ [-\cos t]_0^{\frac{\pi}{2}}$ $=2$	$\frac{1}{2}$ 1 $\frac{1}{2}$
24.	If $\vec{a}$ and $\vec{b}$ are two non-zero vectors such that $(\vec{a}+\vec{b})\perp\vec{a}$ and $(2\vec{a}+\vec{b})\perp\vec{b}$, then prove that $\lvert\vec{b}\rvert=\sqrt{2}\lvert\vec{a}\rvert$.	
Sol.	$(\vec{a}+\vec{b}).\vec{a}=0\ \Rightarrow\ \lvert\vec{a}\rvert^2+\vec{b}.\vec{a}=0$ ----------(1) $(2\vec{a}+\vec{b}).\vec{b}=0\ \Rightarrow 2\vec{a}.\vec{b}+\lvert\vec{b}\rvert^2=0$ ----------(2) $2\,(-\lvert\vec{a}\rvert^2)+\lvert\vec{b}\rvert^2=0$ {Using (1) and (2)} $\lvert\vec{b}\rvert^2=2\lvert\vec{a}\rvert^2\Rightarrow\ \lvert\vec{b}\rvert=\sqrt{2}\lvert\vec{a}\rvert$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

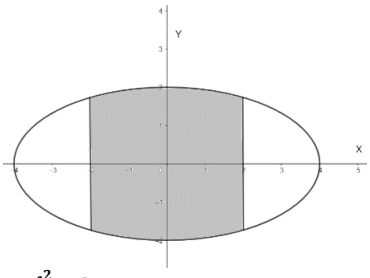
25.	In the given figure, ABCD is a parallelogram. If $\overrightarrow{AB} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\overrightarrow{DB} = 3\hat{i} - 6\hat{j} + 2\hat{k}$, then find $\overrightarrow{AD}$ and hence find the area of parallelogram ABCD.  	
Sol.	$\overrightarrow{AD} + \overrightarrow{DB} = \overrightarrow{AB}$ $\overrightarrow{AD} = (2\hat{\imath} - 4\hat{\jmath} + 5\hat{k}) - (3\hat{\imath} - 6\hat{\jmath} + 2\hat{k})$ $= -\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ $\overrightarrow{AD} \times \overrightarrow{AB} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -1 & 2 & 3 \\ 2 & -4 & 5 \end{vmatrix} = 22\hat{\imath} + 11\hat{\jmath}$ Area $= \lvert\overrightarrow{AD} \times \overrightarrow{AB}\rvert = \lvert 22\hat{\imath} + 11\hat{\jmath}\rvert$ $= \sqrt{605}$ *or* $11\sqrt{5}$	$\frac{1}{2}$ 1 $\frac{1}{2}$
	SECTION C **In this section there are 6 short answer type questions of 3 marks each.**	
26(a).	A relation R on set A = {1, 2, 3, 4, 5} is defined as R = {(x, y) : $\lvert x^2 - y^2\rvert < 8$}. Check whether the relation R is reflexive, symmetric and transitive.	
Sol.	**(a) Reflexive:** $\because \lvert x^2 - x^2\rvert < 8 \; \forall \; x \in A \Rightarrow (x, x) \in R \;\; \therefore R$ **is reflexive .** **(b) Symmetric:** **Let (x,y)** $\in R$ **for some x,y** $\in$ **A** $\therefore \lvert x^2 - y^2\rvert < 8 \Rightarrow \lvert y^2 - x^2\rvert < 8 \Rightarrow (y, x) \in R$ **Hence R is symmetric.** **(c) Transitive:** **(1,2) , (2,3)** $\in R$ **as** $\lvert 1^2 - 2^2\rvert < 8$, $\lvert 2^2 - 3^2\rvert < 8$ **respectively** **But** $\lvert 1^2 - 3^2\rvert \not< 8 \Rightarrow (1, 3) \notin R$ **Hence** R **is not transitive.**	$\frac{1}{2}$ 1 $1\frac{1}{2}$
	OR	
26(b).	A function f is defined from R → R as f(x) = ax + b, such that f(1) = 1 and f(2) = 3. Find function f(x). Hence, check whether function f(x) is one-one and onto or not.	

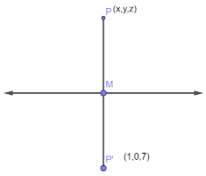
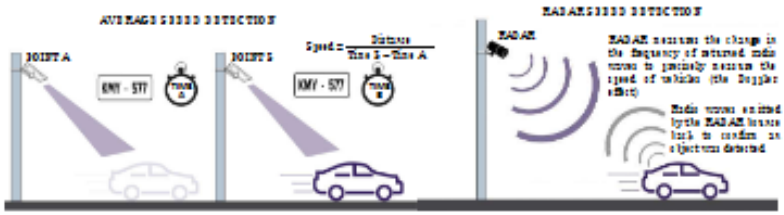
Sol.	$f(x) = ax + b$ Solving $a+ b = 1$ and $2a + b = 3$ to get $a=2$, $b = -1$ $f(x) = 2x - 1$ Let $f(x_1) = f(x_2)$ for some $x_1, x_2 \in R$ $2x_1 - 1 = 2x_2 - 1 \Rightarrow x_1 = x_2$ Hence f is one – one. Let $y = 2x - 1$, $y \in R$ (Codomain) $\Rightarrow x = \frac{y+1}{2} \in R$ (domain) Also, $f(x) = f(\frac{y+1}{2}) = y$ $\therefore$ f is onto.	1 1 1
27(a).	If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.	
Sol.	$\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ Put $x = \sin\theta$, $y = \sin\phi$ $\Rightarrow \cos\theta + \cos\phi = a(\sin\theta - \sin\phi)$ $\Rightarrow 2\cos(\frac{\theta+\phi}{2})\cos(\frac{\theta-\phi}{2}) = 2a\sin(\frac{\theta-\phi}{2})\cos(\frac{\theta+\phi}{2})$ $\Rightarrow \cot(\frac{\theta-\phi}{2}) = a$ $\Rightarrow \theta - \phi = 2\cot^{-1}a$ $\Rightarrow \sin^{-1}x - \sin^{-1}y = 2\cot^{-1}a$ $\Rightarrow \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0$ $\Rightarrow \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$
	OR	
27(b).	If $y = (\tan x)^x$, then find $\frac{dy}{dx}$.	
Sol.	$y = (\tan x)^x$ $\log y = x\log(\tan x)$ $\frac{1}{y}\frac{dy}{dx} = x\left(\frac{sec^2x}{\tan x}\right) + \log(\tan x)$ $\frac{dy}{dx} = (\tan x)^x\left[\left(\frac{x\,sec^2x}{\tan x}\right) + \log(\tan x)\right]$	$\frac{1}{2}$ 2 $\frac{1}{2}$
28(a).	Find : $\int \frac{x^2}{(x^2+4)(x^2+9)}\,dx$	

Sol.	Let $I = \int \frac{x^2}{(x^2+4)(x^2+9)}dx$ Put $x^2 = t$ $\frac{t}{(t+4)(t+9)} = \frac{A}{t+4} + \frac{B}{t+9} \Rightarrow A = \frac{-4}{5}, B = \frac{9}{5}$ $I = \frac{-4}{5}\int \frac{1}{2^2+x^2}dx + \frac{9}{5}\int \frac{1}{3^2+x^2}dx$ $= \frac{-2}{5}\tan^{-1}\left(\frac{x}{2}\right) + \frac{3}{5}\tan^{-1}\left(\frac{x}{3}\right) + C$	$\frac{1}{2}$ $1\frac{1}{2}$ 1
	OR	
28(b).	Evaluate : $\int_1^3 (\lvert x-1\rvert + \lvert x-2\rvert + \lvert x-3\rvert)\,dx$	
Sol.	$\int_1^3 (\lvert x-1\rvert + \lvert x-2\rvert + \lvert x-3\rvert)dx$ $= \int_1^3 (x-1)dx + \int_1^2 -(x-2)dx + \int_2^3 (x-2)dx - \int_1^3 (x-3)dx$ $= \int_1^3 2\,dx + \int_1^2 (2-x)\,dx + \int_2^3 (x-2)dx$ $= [2x]_1^3 + \left[\frac{(2-x)^2}{-2}\right]_1^2 + \left[\frac{(x-2)^2}{2}\right]_2^3$ $= 4 + \frac{1}{2} + \frac{1}{2} = 5$	$1\frac{1}{2}$ $1\frac{1}{2}$
29.	Find the particular solution of the differential equation given by $x^2\frac{dy}{dx} - xy = x^2\cos^2\left(\frac{y}{2x}\right)$, given that when $x = 1, y = \frac{\pi}{2}$.	
Sol.	$\frac{dy}{dx} = \frac{y}{x} + \cos^2\left(\frac{y}{2x}\right)$ Put $y = vx$ so that $\frac{dy}{dx} = v + x\frac{dv}{dx}$ $\Rightarrow v + x\frac{dv}{dx} = v + \cos^2\left(\frac{v}{2}\right)$ $\Rightarrow \int \sec^2\left(\frac{v}{2}\right)dv = \int \frac{1}{x}dx$ $\Rightarrow 2\tan\left(\frac{v}{2}\right) = \log\lvert x\rvert + C$ $\Rightarrow 2\tan\left(\frac{y}{2x}\right) = \log\lvert x\rvert + C$ $2\tan\frac{\pi}{4} = \log 1 + C \Rightarrow C = 2 \Rightarrow 2\tan\left(\frac{y}{2x}\right) = \log\lvert x\rvert + 2$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$

30.	Solve the following linear programming problem graphically : Maximise z = 500x + 300y, subject to constraints $x + 2y \leq 12$ $2x + y \leq 12$ $4x + 5y \geq 20$ $x \geq 0, y \geq 0$	
Sol.	**Max z = 500x + 300y**  **Corner Point / Z** **A (0,4) — 1200** **B (0,6) — 1800** **C (4,4) — 3200** **D (6,0) — 3000** **E (5,0) — 2500** **Max z = 3200 at x = 4, y = 4**	**Correct Graph - $1\frac{1}{2}$** **Correct Table - 1** $\frac{1}{2}$
31.	E and F are two independent events such that $P(\overline{E}) = 0.6$ and $P(E \cup F) = 0.6$. Find P(F) and $P(\overline{E} \cup \overline{F})$.	
Sol.	$P(\overline{E}) = 0.6 \Rightarrow P(E) = 0.4$ $P(E \cup F) = P(E) + P(F) - P(E \cap F)$ $\Rightarrow 0.6 = 0.4 + P(F) - 0.4\,P(F) \Rightarrow P(F) = \frac{1}{3}$ $P(\overline{E} \cup \overline{F}) = 1 - P(E \cap F)$ $= 1 - 0.4 \times \frac{1}{3} = \frac{13}{15}$	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$
	SECTION D **In this section there are 4 long answer type questions of 5 marks each.**	

32(a).	If $A=\begin{bmatrix}1 & -2 & 0\\ 2 & -1 & -1\\ 0 & -2 & 1\end{bmatrix}$, find A^{-1} and use it to solve the following system of equations : $x-2y=10,\ 2x-y-z=8,\ -2y+z=7$	
Sol.	$\|A\|=1\neq 0$ hence A^{-1} exists. $Adj\ A=\begin{bmatrix}-3 & 2 & 2\\ -2 & 1 & 1\\ -4 & 2 & 3\end{bmatrix}$ $A^{-1}=\begin{bmatrix}-3 & 2 & 2\\ -2 & 1 & 1\\ -4 & 2 & 3\end{bmatrix}$ $AX=B\Rightarrow\begin{bmatrix}1 & -2 & 0\\ 2 & -1 & -1\\ 0 & -2 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}10\\ 8\\ 7\end{bmatrix}$ $X=A^{-1}B\Rightarrow\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-3 & 2 & 2\\ -2 & 1 & 1\\ -4 & 2 & 3\end{bmatrix}\begin{bmatrix}10\\ 8\\ 7\end{bmatrix}=\begin{bmatrix}0\\ -5\\ -3\end{bmatrix}$ $\Rightarrow x=0, y=-5, z=-3$	1 2 $\frac{1}{2}$ $1\frac{1}{2}$
	OR	
32(b).	If $A=\begin{bmatrix}-1 & a & 2\\ 1 & 2 & x\\ 3 & 1 & 1\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}1 & -1 & 1\\ -8 & 7 & -5\\ b & y & 3\end{bmatrix}$, find the value of $(a+x)-(b+y)$.	
Sol.	$AA^{-1}=I$ $\begin{bmatrix}-1 & a & 2\\ 1 & 2 & x\\ 3 & 1 & 1\end{bmatrix}\begin{bmatrix}1 & -1 & 1\\ -8 & 7 & -5\\ b & y & 3\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ $\begin{bmatrix}-1-8a+2b & 1+7a+2y & 5-5a\\ -15+bx & 13+xy & 3x-9\\ -5+b & 4+y & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ $-5+b=0\Rightarrow b=5,\quad 5-5a=0\Rightarrow a=1$ $4+y=0\Rightarrow y=-4,\quad 3x-9=0\Rightarrow x=3$ $\therefore (a+x)-(b+y)=(1+3)-(5-4)=3$	1 $1\frac{1}{2}$ 1 1 $\frac{1}{2}$
33(a).	Evaluate : $\int_0^{\frac{\pi}{4}}\frac{\sin x+\cos x}{9+16\sin 2x}\,dx$	
Sol.	Let $I=\int_0^{\frac{\pi}{4}}\frac{\sin x+\cos x}{9+16\sin 2x}dx$	

	Put $\sin x - \cos x = t$, so that $(\cos x + \sin x)\,dx = dt$ $sin^2x + cos^2x - sin2x = t^2 \Rightarrow sin2x = 1 - t^2$ $I = \int_{-1}^{0} \frac{dt}{25-16t^2}$ $= \frac{1}{16}\int_{-1}^{0} \frac{dt}{(\frac{5}{4})^2 - t^2}$ $= \frac{1}{40}\left[log\left\|\frac{5+4t}{5-4t}\right\|\right]_{-1}^{0}$ $= \frac{1}{40}\left[\log 1 - \log\left(\frac{1}{9}\right)\right] = \frac{1}{40}\log 9$ *or* $\frac{1}{20}\log 3$	1 $\frac{1}{2}$ 1 $1\frac{1}{2}$ 1
	OR	
33(b).	Evaluate : $\int_0^{\frac{\pi}{2}} \sin 2x \tan^{-1}(\sin x)\,dx$	
Sol.	*Let* $I = \int_0^{\frac{\pi}{2}} \sin 2x \tan^{-1}(sin\,x)\,dx$ Put $\sin x = t$ so that $\cos x\,dx = dt$ $I = 2\int_0^1 t \tan^{-1} t\,dt$ $= 2\left[\tan^{-1} t \left(\frac{t^2}{2}\right) - \frac{1}{2}\int \frac{t^2}{1+t^2} dt\right]_0^1$ $= 2\left[\left(\frac{t^2}{2}\right)\tan^{-1} t - \frac{1}{2}t + \frac{1}{2}\tan^{-1} t\right]_0^1$ $= 2\left(\frac{\pi}{4} - \frac{1}{2}\right) = \frac{\pi}{2} - 1$	1 1 $1\frac{1}{2}$ 1 $\frac{1}{2}$
34.	Using integration, find the area of the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$, included between the lines $x = -2$ and $x = 2$.	
Sol.	 Area $= 4\int_0^2 y\,dx$ $= 4\left[\frac{1}{2}\int_0^2 \sqrt{4^2 - x^2}\,dx\right]$ $= 2\left[\frac{x}{2}\sqrt{4^2 - x^2} + 8\sin^{-1}\left(\frac{x}{4}\right)\right]_0^2$ $= 2\left[\sqrt{12} + \frac{8\pi}{6}\right] = 4\sqrt{3} + \frac{8\pi}{3}$	Correct graph-1 1 2 1

35.	The image of point P(x, y, z) with respect to line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is P′ (1, 0, 7). Find the coordinates of point P.	
Sol.	Let foot of the perpendicular on the given line from point P be M $(\lambda, 2\lambda+1, 3\lambda+2)$ D. ratios of PP' are $\lambda-1,\ 2\lambda+1,\ 3\lambda-5$ $1(\lambda\text{-}1) + 2(2\lambda+1) + 3(3\lambda\text{-}5) = 0$ $\Rightarrow\ \lambda = 1$ Coordinates of M(1,3,5) $\frac{x+1}{2} = 1,\ \frac{y+0}{2} = 3,\ \frac{z+7}{2} = 5$ $\Rightarrow x = 1,\ y = 6,\ z = 3\ \Rightarrow P(1,6,3)$ 	1 1 1 1 1
	SECTION E In this section there are 3 case-study based questions of 4 marks each.	
36.	The traffic police has installed Over Speed Violation Detection (OSVD) system at various locations in a city. These cameras can capture a speeding vehicle from a distance of 300 m and even function in the dark.  A camera is installed on a pole at the height of 5 m. It detects a car travelling away from the pole at the speed of 20 m/s. At any point, x m away from the base of the pole, the angle of elevation of the speed camera from the car C is θ. On the basis of the above information, answer the following questions : (i) Express θ in terms of height of the camera installed on the pole and x. 1 (ii) Find $\frac{d\theta}{dx}$. 1 (iii) (a) Find the rate of change of angle of elevation with respect to time at an instant when the car is 50 m away from the pole. 2 **OR** (iii) (b) If the rate of change of angle of elevation with respect to time of another car at a distance of 50 m from the base of the pole is $\frac{3}{101}$ rad/s, then find the speed of the car. 2	
Sol.	(i) $\tan\theta = \frac{5}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{5}{x}\right)$ (ii) $\frac{d\theta}{dx} = \frac{-5}{5^2+x^2}$	1 1

Marking Scheme
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MATHEMATICS PAPER CODE 65/2/1

1.	If the sum of all the elements of a 3×3 scalar matrix is 9, then the product of all its elements is : (A) 0 (B) 9 (C) 27 (D) 729	
Sol.	(A) 0	1
2.	Let $f : R_+ \to [-5, \infty)$ be defined as $f(x) = 9x^2 + 6x - 5$, where R_+ is the set of all non-negative real numbers. Then, f is : (A) one-one (B) onto (C) bijective (D) neither one-one nor onto	
Sol.	(C) Bijective	1
3.	If $\begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix} = kabc$, then the value of k is : (A) 0 (B) 1 (C) 2 (D) 4	
Sol.	(D) 4	1
4.	The number of points of discontinuity of $f(x) = \begin{cases} \lvert x\rvert + 3, & \text{if } x \le -3 \\ -2x, & \text{if } -3 < x < 3 \\ 6x + 2, & \text{if } x \ge 3 \end{cases}$ is : (A) 0 (B) 1 (C) 2 (D) infinite	
Sol.	(B) 1	1

5.	The function $f(x) = x^3 - 3x^2 + 12x - 18$ is : (A) strictly decreasing on R (B) strictly increasing on R (C) neither strictly increasing nor strictly decreasing on R (D) strictly decreasing on $(-\infty, 0)$	
Sol.	(B) Strictly increasing on R	1
6.	$\int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x \cos x} dx$ is equal to : (A) π (B) Zero (0) (C) $\int_0^{\pi/2} \frac{2\sin x}{1+\sin x \cos x} dx$ (D) $\frac{\pi^2}{4}$	
Sol.	(B) Zero (0)	1
7.	The differential equation $\frac{dy}{dx} = F(x, y)$ will not be a homogeneous differential equation, if F(x, y) is : (A) $\cos x - \sin\left(\frac{y}{x}\right)$ (B) $\frac{y}{x}$ (C) $\frac{x^2+y^2}{xy}$ (D) $\cos^2\left(\frac{x}{y}\right)$	
Sol.	(A) $cosx - sin\left(\frac{y}{x}\right)$	1
8.	For any two vectors $\vec{a}$ and $\vec{b}$, which of the following statements is always true ? (A) $\vec{a}.\vec{b} \geq \lvert\vec{a}\rvert\lvert\vec{b}\rvert$ (B) $\vec{a}.\vec{b} = \lvert\vec{a}\rvert\lvert\vec{b}\rvert$ (C) $\vec{a}.\vec{b} \leq \lvert\vec{a}\rvert\lvert\vec{b}\rvert$ (D) $\vec{a}.\vec{b} < \lvert\vec{a}\rvert\lvert\vec{b}\rvert$	
Sol.	(C) $\vec{a}.\vec{b} \leqslant \lvert\vec{a}\rvert\lvert\vec{b}\rvert$	1

9.	The coordinates of the foot of the perpendicular drawn from the point (0, 1, 2) on the x-axis are given by : (A) (1, 0, 0) (B) (2, 0, 0) (C) $(\sqrt{5}, 0, 0)$ (D) (0, 0, 0)	
Sol.	(D) (0, 0, 0)	1
10.	The common region determined by all the constraints of a linear programming problem is called : (A) an unbounded region (B) an optimal region (C) a bounded region (D) a feasible region	
Sol.	(D) a feasible region	1
11.	Let E be an event of a sample space S of an experiment, then P(S | E) = (A) $P(S \cap E)$ (B) P(E) (C) 1 (D) 0	
Sol.	(C) 1	1
12.	If A = $[a_{ij}]$ be a 3×3 matrix, where $a_{ij} = i - 3j$, then which of the following is ***false*** ? (A) $a_{11} < 0$ (B) $a_{12} + a_{21} = -6$ (C) $a_{13} > a_{31}$ (D) $a_{31} = 0$	
Sol.	(C) $a_{13} > a_{31}$	1
13.	The derivative of $\tan^{-1}(x^2)$ w.r.t. x is : (A) $\frac{x}{1+x^4}$ (B) $\frac{2x}{1+x^4}$ (C) $-\frac{2x}{1+x^4}$ (D) $\frac{1}{1+x^4}$	
Sol.	(B) $\frac{2x}{1+x^4}$	1

14.	The degree of the differential equation $(y'')^2 + (y')^3 = x \sin(y')$ is : (A) 1 (B) 2 (C) 3 (D) not defined	
Sol.	(D) not defined	1
15.	The unit vector perpendicular to both vectors $\hat{i} + \hat{k}$ and $\hat{i} - \hat{k}$ is : (A) $2\hat{j}$ (B) $\hat{j}$ (C) $\frac{\hat{i} - \hat{k}}{\sqrt{2}}$ (D) $\frac{\hat{i} + \hat{k}}{\sqrt{2}}$	
Sol.	(B) $\hat{j}$	1
16.	Direction ratios of a vector parallel to line $\frac{x-1}{2} = -y = \frac{2z+1}{6}$ are : (A) 2, – 1, 6 (B) 2, 1, 6 (C) 2, 1, 3 (D) 2, – 1, 3	
Sol.	(D) 2, -1, 3	1
17.	If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $[F(x)]^2 = F(kx)$, then the value of k is : (A) 1 (B) 2 (C) 0 (D) – 2	
Sol.	(B) 2	1
18.	If a line makes an angle of 30° with the positive direction of x-axis, 120° with the positive direction of y-axis, then the angle which it makes with the positive direction of z-axis is : (A) 90° (B) 120° (C) 60° (D) 0°	
Sol.	(A) 90^0	1

	*Questions number **19** and **20** are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.* (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	
19.	*Assertion (A) :* For any symmetric matrix A, B′AB is a skew-symmetric matrix. *Reason (R) :* A square matrix P is skew-symmetric if P′ = – P.	
Sol.	(D) Assertion (A) is false, but Reason (R) is true	1
20.	*Assertion (A) :* For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. *Reason (R) :* For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$.	
Sol.	(C) Assertion (A) is true, but Reason (R) is false.	1
	SECTION B	
21.	(a) Find the value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$.	
Sol.	The given expression $= \frac{-\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4}$ $= \frac{-\pi}{12}$	$1\frac{1}{2}$ $\frac{1}{2}$
	OR	
	(b) Find the domain of the function $f(x) = \sin^{-1}(x^2 - 4)$. Also, find its range.	
Sol.	$-1 \leqslant x^2 - 4 \leqslant 1$ $\Rightarrow 3 \leqslant x^2 \leqslant 5$ Domain $= [-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$	$\frac{1}{2}$ $\frac{1}{2}$

	Range = $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	1
22.	(a) If f(x) = $\lvert \tan 2x \rvert$, then find the value of f′(x) at x = $\frac{\pi}{3}$.	
Sol.	$f(x) = -tan2x,\ \frac{\pi}{4} < x < \frac{\pi}{2}$ $f'(x) = -2sec^2 2x,\ \frac{\pi}{4} < x < \frac{\pi}{2}$ $f'(\frac{\pi}{3}) = -2(-2)^2 = -8$	$\frac{1}{2}$ 1 $\frac{1}{2}$
	OR	
	(b) If y = cosec(cot^{-1} x), then prove that $\sqrt{1+x^2}\,\frac{dy}{dx} - x = 0$.	
Sol.	$y = \sqrt{1 + cot^2(cot^{-1}x)} = \sqrt{1+x^2}$ $\Rightarrow \frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}$ $\Rightarrow \sqrt{1+x^2}\frac{dy}{dx} - x = 0$	$\frac{1}{2}$ 1 $\frac{1}{2}$
23.	If M and m denote the local maximum and local minimum values of the function f(x) = x + $\frac{1}{x}$ (x ≠ 0) respectively, find the value of (M – m).	
Sol.	$f'(x) = 1 - \frac{1}{x^2} = \frac{x^2-1}{x^2} = \frac{(x+1)(x-1)}{x^2}$ $f'(x) = 0 \Rightarrow x = -1,\ 1$ $f''(x) = \frac{2}{x^3} \Rightarrow f''(-1) = -2 < 0$ ∴ -1 is a point of local maximum The local maximum value = $f(-1) = -2 = M$ $f''(1) = 2 > 0$ ∴ 1 is point of local minimum The local minimum value = $f(1) = 2 = m$ $M - m = -4$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

24.	**Find :** $\int \frac{e^{4x}-1}{e^{4x}+1}\,dx$	
Sol.	Given integral $=\int \frac{e^{4x}-1}{e^{4x}+1}dx$ $=\int \frac{e^{2x}-e^{-2x}}{e^{2x}+e^{-2x}}dx$ $=\frac{1}{2}\int \frac{2(e^{2x}-e^{-2x})}{e^{2x}+e^{-2x}}dx$ $=\frac{1}{2}log\lvert e^{2x}+e^{-2x}\rvert + C$	$\frac{1}{2}$ $\frac{1}{2}$ 1
25.	Show that $f(x) = e^x - e^{-x} + x - \tan^{-1} x$ is strictly increasing in its domain.	
Sol.	$f'(x) = e^x + e^{-x} + 1 - \frac{1}{1+x^2}$ $=e^x + \frac{1}{e^x} + \frac{x^2}{1+x^2} > 0\ for\ all\ x \in R$ $\therefore$ f is strictly increasing over its domain R	1 1
	Section C	
26.	(a) If $x = e^{\cos 3t}$ and $y = e^{\sin 3t}$, prove that $\frac{dy}{dx} = -\frac{y \log x}{x \log y}$.	
Sol	$\frac{dx}{dt} = e^{cos3t} \times (-sin3t) \times 3$ $\frac{dy}{dt} = e^{sin3t} \times (cos3t) \times 3$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e^{sin3t} \times (cos3t)}{-e^{cos3t} \times (sin3t)}$ $x = e^{cos3t} \Longrightarrow cos3t = logx$ $y = e^{sin3t} \Longrightarrow sin3t = logy$ $\therefore \frac{dy}{dx} = \frac{-ylogx}{xlogy}$	1 1 1

	OR	
	(b) Show that : $\frac{d}{dx}(\lvert x \rvert) = \frac{x}{\lvert x \rvert}, x \neq 0$	
Sol.	$\frac{d(\lvert x \rvert)}{dx} = \frac{d(\sqrt{x^2})}{dx}, x \neq 0$ $= \frac{1}{2}(x^2)^{-\frac{1}{2}} \times \frac{d(x^2)}{dx}$ $= \frac{1}{2\sqrt{x^2}} 2x = \frac{x}{\lvert x \rvert}$	1 $1\frac{1}{2}$ $\frac{1}{2}$
27.	(a) Evaluate : $\int_{-2}^{2} \sqrt{\frac{2-x}{2+x}}\, dx$	
Sol.	(a) $\int_{-2}^{2} \sqrt{\frac{2-x}{2+x}}\, dx$ $= \int_{-2}^{2} \frac{2-x}{\sqrt{4-x^2}} dx$ $= \int_{-2}^{2} \frac{2}{\sqrt{4-x^2}} dx - \int_{-2}^{2} \frac{x}{\sqrt{4-x^2}} dx$ $= 2\int_{0}^{2} \frac{2}{\sqrt{4-x^2}} dx - 0$ [$\frac{2}{\sqrt{4-x^2}}$ is even, $\frac{x}{\sqrt{4-x^2}}$ is odd] $= 4\int_{0}^{2} \frac{1}{\sqrt{4-x^2}} dx$ $= 4\, sin^{-1}\frac{x}{2}\Big\rvert_{0}^{2}$ $= 2\pi$	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$
	OR	
	(b) Find : $\int \frac{1}{x\,[(\log x)^2 - 3\log x - 4]} dx$	
Sol.	Let $logx = t \Rightarrow \frac{1}{x} dx = dt$ The given integral becomes $= \int \frac{1}{t^2-3t-4} dt$ $= \int \frac{1}{\left(t-\frac{3}{2}\right)^2 - \left(\frac{5}{2}\right)^2} dx$	$\frac{1}{2}$ $\frac{1}{2}$ 1

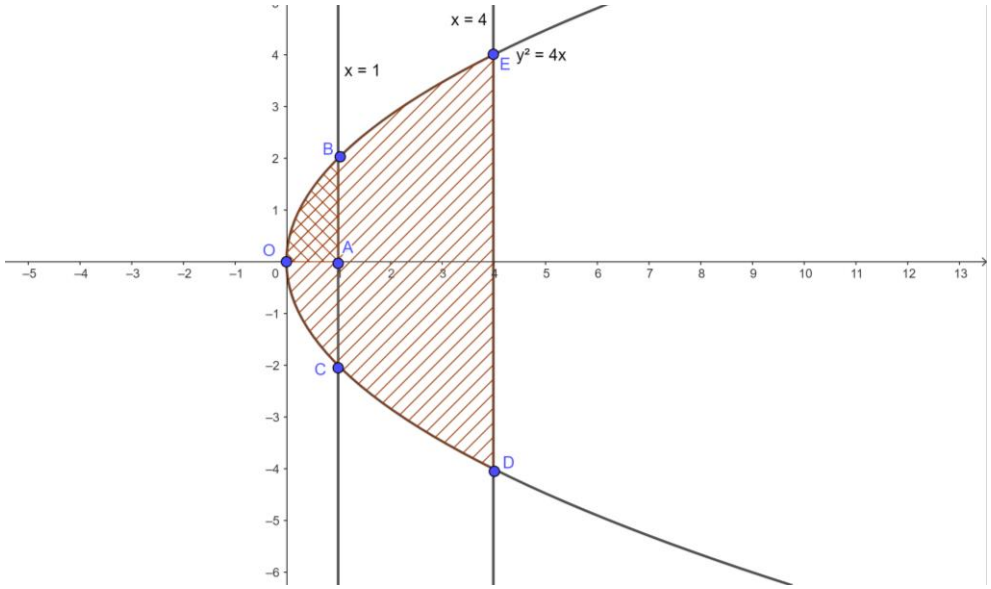
	$= \frac{1}{5} log \left\lvert \frac{t-4}{t+1} \right\rvert + C$ $= \frac{1}{5} log \left\lvert \frac{logx-4}{logx+1} \right\rvert + C$	$\frac{1}{2}$ $\frac{1}{2}$
28.	(a) Find the particular solution of the differential equation given by $2xy + y^2 - 2x^2 \frac{dy}{dx} = 0$; y = 2, when x = 1.	
Sol.	Given differential equation can be written as $\frac{dy}{dx} = \frac{2xy+y^2}{2x^2} = \frac{y}{x} + \frac{y^2}{2x^2}$ Let $y = vx \Longrightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ The equation becomes $x\frac{dv}{dx} = \frac{1}{2}v^2$ $\Longrightarrow \frac{dv}{v^2} = \frac{1}{2} \times \frac{dx}{x}$ Integrating both sides, we get $\frac{-1}{v} = \frac{1}{2} log\lvert x \rvert + C$ $\Longrightarrow -\frac{x}{y} = \frac{1}{2} log\lvert x \rvert + C$ $y = 2, x = 1$ gives $C = -\frac{1}{2}$ The particular solution is $-\frac{x}{y} = \frac{1}{2} log\lvert x \rvert - \frac{1}{2}$ $or, y = \frac{2x}{1 - log\lvert x \rvert}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$
	OR	
	(b) Find the general solution of the differential equation : $y\,dx = (x + 2y^2)\,dy$	
Sol.	Given differential equation can be written as $\frac{dx}{dy} - \frac{x}{y} = 2y$ Integrating Factor = $e^{\int \frac{-1}{y} dy} = \frac{1}{y}$ Solution is $x\frac{1}{y} = \int 2dy$ $\Longrightarrow \frac{x}{y} = 2y + C$ $\Longrightarrow x = 2y^2 + Cy$	1 1 $\frac{1}{2}$ $\frac{1}{2}$

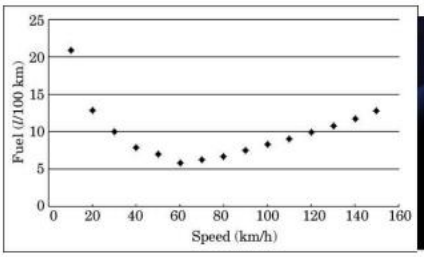
29.	The position vectors of vertices of Δ ABC are A($2\hat{i} - \hat{j} + \hat{k}$), B($\hat{i} - 3\hat{j} - 5\hat{k}$) and C($3\hat{i} - 4\hat{j} - 4\hat{k}$). Find all the angles of Δ ABC.	
Sol.	$cosA = \frac{\overrightarrow{AB}.\overrightarrow{AC}}{\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{AC}\rvert} = \frac{(-\hat{i} - 2\hat{j} - 6\hat{k}).(\hat{i} - 3\hat{j} - 5\hat{k})}{\sqrt{41}\sqrt{35}} = \frac{35}{\sqrt{41}\sqrt{35}} = \frac{\sqrt{35}}{\sqrt{41}}$ $A = cos^{-1}\left(\frac{\sqrt{35}}{\sqrt{41}}\right)$ $cosB = \frac{\overrightarrow{BA}.\overrightarrow{BC}}{\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert} = \frac{(\hat{i} + 2\hat{j} + 6\hat{k}).(2\hat{i} - \hat{j} + \hat{k})}{\sqrt{41}\sqrt{6}} = \frac{6}{\sqrt{41}\sqrt{6}} = \frac{\sqrt{6}}{\sqrt{41}}$ $B = cos^{-1}\left(\frac{\sqrt{6}}{\sqrt{41}}\right)$ $cosC = \frac{\overrightarrow{CB}.\overrightarrow{CA}}{\lvert\overrightarrow{CB}\rvert\lvert\overrightarrow{CA}\rvert} = \frac{(-2\hat{i} + \hat{j} - \hat{k}).(-\hat{i} + 3\hat{j} + 5\hat{k})}{\lvert\overrightarrow{CB}\rvert\lvert\overrightarrow{CA}\rvert} = 0$ cos C = 0 $\Rightarrow C = \frac{\pi}{2}$	1 1 1
30.	A pair of dice is thrown simultaneously. If X denotes the absolute difference of the numbers appearing on top of the dice, then find the probability distribution of X.	
Sol.	<table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>P(X)</td><td>$\frac{6}{36} = \frac{1}{6}$</td><td>$\frac{10}{36} = \frac{5}{18}$</td><td>$\frac{8}{36} = \frac{2}{9}$</td><td>$\frac{6}{36} = \frac{1}{6}$</td><td>$\frac{4}{36} = \frac{1}{9}$</td><td>$\frac{2}{36} = \frac{1}{18}$</td></tr></table>	$\frac{1}{2} \times 6 = 3$
31.	Find : $\int x^2 \cdot \sin^{-1}(x^{3/2})\, dx$	
Sol.	Let $x^{\frac{3}{2}} = t$ $\Rightarrow \frac{3}{2}x^{\frac{1}{2}}dx = dt$ The given integral becomes $\frac{2}{3}\int t\, sin^{-1}t dt$ $= \frac{2}{3}\left[sin^{-1}t \times \frac{t^2}{2} - \int \frac{1}{\sqrt{1-t^2}} \times \frac{t^2}{2} dt\right]$ $= \frac{1}{3}\left[sin^{-1}t \times t^2 + \int \frac{1-t^2-1}{\sqrt{1-t^2}} dt\right]$ $= \frac{1}{3}\left[sin^{-1}t \times t^2 + \int \sqrt{1-t^2}\, dt - \int \frac{1}{\sqrt{1-t^2}} dt\right]$ $= \frac{1}{3}\left[t^2 sin^{-1}t + \frac{t}{2}\sqrt{1-t^2} + \frac{1}{2}sin^{-1}t - sin^{-1}t\right]$+C	$\frac{1}{2}$ $\frac{1}{2}$ 1

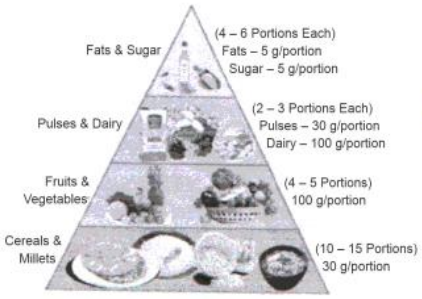
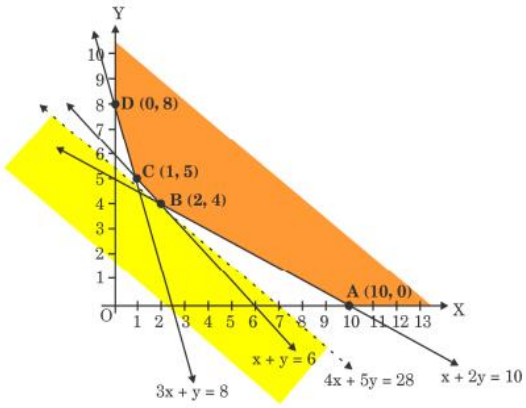
	$=\frac{1}{3}\left[t^2 sin^{-1}t + \frac{t}{2}\sqrt{1-t^2} - \frac{1}{2}sin^{-1}t\right] + C$ $= \frac{1}{3}\left[x^3 sin^{-1}\left(x^{\frac{3}{2}}\right) + \frac{x^{\frac{3}{2}}}{2}\sqrt{1-x^3} - \frac{1}{2}sin^{-1}\left(x^{\frac{3}{2}}\right)\right] + C$	1
	Section D	
32.	(a) Show that a function f : R → R defined by f(x) = $\frac{2x}{1+x^2}$ is neither one-one nor onto. Further, find set A so that the given function f : R → A becomes an onto function.	
Sol.	Let $f(x_1) = f(x_2)\ for\ some\ x_1, x_2 \in R$ Then $\frac{2x_1}{1+{x_1}^2} = \frac{2x_2}{1+{x_2}^2}$ $\Rightarrow x_1 + x_1{x_2}^2 = x_2 + {x_1}^2 x_2$ $\Rightarrow (x_1 - x_2) - x_1x_2(x_1 - x_2) = 0$ $\Rightarrow (x_1 - x_2)(1 - x_1x_2) = 0$ $\Rightarrow x_1 - x_2 = 0$ or $1 - x_1x_2 = 0$ $\Rightarrow x_1 = x_2$ or $x_1x_2 = 1$, so if $x_1x_2 = 1, x_1 \neq x_2$ Hence f is not one -one Let y = f(x) where $x \in R$ Then $y = \frac{2x}{1+x^2}$. Here, for x = 0, y = 0 If $y \neq 0$, then $y = \frac{2x}{1+x^2}$ $\Rightarrow yx^2 - 2x + y = 0$ $\Rightarrow x = \frac{2 \pm \sqrt{4-4y^2}}{2y}$ For x to be real, $4 - 4y^2 \geq 0$ $\Rightarrow y^2 \leq 1$ $\Rightarrow -1 \leq y \leq 1$ Hence, range = $[-1,1] \neq codomain$ Hence, f is not onto. For the given function to become onto, $A = [-1,1]$	2 2 1

	OR	
	(b) A relation R is defined on N × N (where N is the set of natural numbers) as : (a, b) R (c, d) ⇔ a – c = b – d Show that R is an equivalence relation.	
Sol.	Let $(a, b) \in N \times N$ We have $a - a = b - b$ This implies that (a, b) R (a, b) $\forall (a, b) \in N \times N$ Hence R is reflexive Let (a, b) R (c, d) for some $(a, b), (c, d) \in N \times N$ Then $a - c = b - d$ $\Rightarrow c - a = d - b$ $\Rightarrow$(c, d) R (a, b) Hence, R is symmetric. Let (a, b) R (c, d), (c, d) R (e, f) for some $(a, b), (c, d), (e, f) \in N \times N$ Then $a - c = b - d, c - e = d - f$ $\Rightarrow a - c + c - e = b - d + d - f$ $\Rightarrow a - e = b - f$ $\Rightarrow$(a, b) R (e, f) Hence, R is transitive Thus, R is an equivalence relation.	$1\frac{1}{2}$ $1\frac{1}{2}$ 2
33.	Find the equation of the line which bisects the line segment joining points A(2, 3, 4) and B(4, 5, 8) and is perpendicular to the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.	
Sol	Let direction ratios of the required line be a, b, c. ∵ the required line is perpendicular to both the given lines $\therefore 3a - 16b + 7c = 0$ and $3a + 8b - 5c = 0$ $\Rightarrow \frac{a}{24} = \frac{b}{36} = \frac{c}{72}$ $\Rightarrow \frac{a}{2} = \frac{b}{3} = \frac{c}{6}$ The mid-point of the line-segment AB is (3, 4, 6) Hence, the required equation of the line is $\frac{x-3}{2} = \frac{y-4}{3} = \frac{z-6}{6}$	1 $\frac{1}{2}$ $1\frac{1}{2}$ 1 1

34.	(a) Solve the following system of equations, using matrices : $\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=4,\ \frac{4}{x}-\frac{6}{y}+\frac{5}{z}=1,\ \frac{6}{x}+\frac{9}{y}-\frac{20}{z}=2$ where x, y, z ≠ 0	
Sol.	Given system of linear equations is equivalent to AX = B, where $A=\begin{bmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{bmatrix}, X=\begin{bmatrix}\frac{1}{x}\\ \frac{1}{y}\\ \frac{1}{z}\end{bmatrix}, B=\begin{bmatrix}4\\1\\2\end{bmatrix}$ $\|A\| = 1200 \neq 0$ Cofactors of the elements of A are $A_{11}=75,\ A_{12}=110,\ A_{13}=72$ $A_{21}=150,\ A_{22}=-100,\ A_{23}=0$ $A_{31}=75,\ A_{32}=30,\ A_{33}=-24$ $adjA=\begin{bmatrix}75 & 150 & 75\\ 110 & -100 & 30\\ 72 & 0 & -24\end{bmatrix}$ $A^{-1}=\frac{adjA}{\|A\|}=\frac{1}{1200}\begin{bmatrix}75 & 150 & 75\\ 110 & -100 & 30\\ 72 & 0 & -24\end{bmatrix}$ $X=A^{-1}B=\frac{1}{1200}\begin{bmatrix}75 & 150 & 75\\ 110 & -100 & 30\\ 72 & 0 & -24\end{bmatrix}\begin{bmatrix}4\\1\\2\end{bmatrix}$ $=\frac{1}{1200}\begin{bmatrix}600\\400\\240\end{bmatrix}=\begin{bmatrix}\frac{1}{2}\\ \frac{1}{3}\\ \frac{1}{5}\end{bmatrix}$ ∴x = 2, y = 3, z = 5	$\frac{1}{2}$ $\frac{1}{2}$ 2 $\frac{1}{2}$ 1 $\frac{1}{2}$
	OR	
	(b) If $A=\begin{bmatrix}1 & \cot x\\ -\cot x & 1\end{bmatrix}$, show that $A'A^{-1}=\begin{bmatrix}-\cos 2x & -\sin 2x\\ \sin 2x & -\cos 2x\end{bmatrix}$.	
Sol.	$\|A\| = 1+cot^2x = cosec^2x$ $adjA=\begin{bmatrix}1 & -cotx\\ cotx & 1\end{bmatrix}$	$\frac{1}{2}$ $1\frac{1}{2}$

	$A^{-1} = \frac{adjA}{\|A\|} = \frac{1}{cosec^2x}\begin{bmatrix} 1 & -cotx \\ cotx & 1 \end{bmatrix}$ $A' = \begin{bmatrix} 1 & -cotx \\ cotx & 1 \end{bmatrix}$ $A'A^{-1} = \frac{1}{cosec^2x}\begin{bmatrix} 1-cot^2x & -2cotx \\ 2cotx & 1-cot^2x \end{bmatrix}$ $= \begin{bmatrix} sin^2x - cos^2x & -2sinxcosx \\ 2sinxcosx & sin^2x - cos^2x \end{bmatrix}$ $= \begin{bmatrix} -cos2x & -sin2x \\ sin2x & -cos2x \end{bmatrix}$	1 $\frac{1}{2}$ $1\frac{1}{2}$
35.	If A_1 denotes the area of region bounded by $y^2 = 4x$, $x = 1$ and x-axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x$, $x = 4$, find $A_1 : A_2$.	
Sol.	 A_1 = Area (region OABO) = $\int_0^1 2\sqrt{x}\,dx = 2\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^1 = \frac{4}{3}$ A_2 = Area (region ODEO) = $2\int_0^4 2\sqrt{x}\,dx = 4 \times \frac{2}{3}[2^3] = \frac{64}{3}$ $A_1 : A_2 = \frac{4}{3} : \frac{64}{3} = 1:16$	2 1 1 1

	Section E	
36.	Overspeeding increases fuel consumption and decreases fuel economy as a result of tyre rolling friction and air resistance. While vehicles reach optimal fuel economy at different speeds, fuel mileage usually decreases rapidly at speeds above 80 km/h.   The relation between fuel consumption F (l/100 km) and speed V (km/h) under some constraints is given as $F = \frac{V^2}{500} - \frac{V}{4} + 14$. On the basis of the above information, answer the following questions : (i) Find F, when V = 40 km/h. (ii) Find $\frac{dF}{dV}$. (iii) (a) Find the speed V for which fuel consumption F is minimum. **OR** (iii) (b) Find the quantity of fuel required to travel 600 km at the speed V at which $\frac{dF}{dV} = -0{\cdot}01$.	
Sol.	(i) When V = 40 km/h, F = 36/5 ℓ/100km (ii) $\frac{dF}{dV} = \frac{V}{250} - \frac{1}{4}$ (iii)(a) $\frac{dF}{dV} = 0$ $\Longrightarrow$V = 62.5 km/h $\frac{d^2F}{dV^2} = \frac{1}{250} > 0$ at V = 62.5 km/h Hence, F is minimum when V = 62.5 km/h	1 1 1 1 $\frac{1}{2}$ $\frac{1}{2}$

	OR	
	(iii) (b) $\frac{dF}{dV} = -0.01$ $\Rightarrow \frac{V}{250} - \frac{1}{4} = \frac{-1}{100}$ $\Rightarrow V = 60$ km/h $F = \frac{60^2}{500} - \frac{60}{4} + 14 = 6.2\,\ell/100km$ Quantity of fuel required for 600 km $= 6.2 \times 6 = 37.2\,\ell$	1 $\frac{1}{2}$ $\frac{1}{2}$
37.	The month of September is celebrated as the Rashtriya Poshan Maah across the country. Following a healthy and well-balanced diet is crucial in order to supply the body with the proper nutrients it needs. A balanced diet also keeps us mentally fit and promotes improved level of energy.  Figure-1  Figure-2 A dietician wishes to minimize the cost of a diet involving two types of foods, food X (x kg) and food Y (y kg) which are available at the rate of ₹ 16/kg and ₹ 20/kg respectively. The feasible region satisfying the constraints is shown in Figure-2. On the basis of the above information, answer the following questions : (i) Identify and write all the constraints which determine the given feasible region in Figure-2. (ii) If the objective is to minimize cost Z = 16x + 20y, find the values of x and y at which cost is minimum. Also, find minimum cost assuming that minimum cost is possible for the given unbounded region.	

Sol.

(i)Constraints are $x + 2y \geq 10$ — $1\frac{1}{2}$

$x + y \geq 6$

$3x + y \geq 8$ — $\frac{1}{2}$

$x \geq 0$

$y \geq 0$

(ii)

Corner points	Value of Z = 16x + 20y
A (10, 0)	160
B (2, 4)	112
C (1, 5)	116
D (0, 8)	160

$1\frac{1}{2}$

The minimum cost is ₹112 — $\frac{1}{2}$

38.

Airplanes are by far the safest mode of transportation when the number of transported passengers are measured against personal injuries and fatality totals.

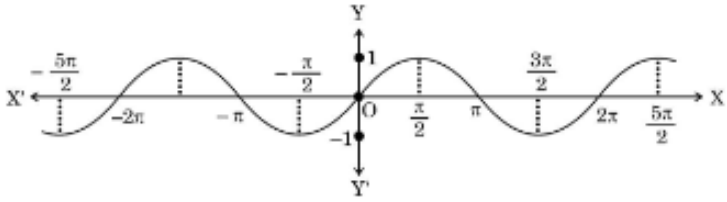


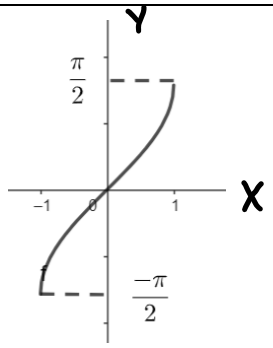


Previous records state that the probability of an airplane crash is 0·00001%. Further, there are 95% chances that there will be survivors after a plane crash. Assume that in case of no crash, all travellers survive.

	Let E_1 be the event that there is a plane crash and E_2 be the event that there is no crash. Let A be the event that passengers survive after the journey. On the basis of the above information, answer the following questions : (i) Find the probability that the airplane will not crash. (ii) Find $P(A \mid E_1) + P(A \mid E_2)$. (iii) (a) Find P(A). **OR** (iii) (b) Find $P(E_2 \mid A)$.	
Sol.	(i) $P(E_2) = 1 - 0.0000001$ $= 0.9999999$ (ii) $P(A/E_1) + P(A/E_2) = \frac{95}{100} + 1 = \frac{195}{100}$	1 1
	(iii)(a) $P(A) = P(E_1) \times P(A/E_1) + P(E_2) \times P(A/E_2)$ $= \frac{1}{10000000} \times \frac{95}{100} + \frac{9999999}{10000000} \times 1$ $= \frac{95+999999900}{1000000000} = \frac{999999995}{1000000000}$	1 1
	OR	
	(iii)(b) $P(E_2/A) = \frac{P(E_2) \times P(A/E_2)}{P(E_1) \times P(A/E_1) + P(E_2) \times P(A/E_2)}$ $= \frac{\frac{9999999}{10000000}}{\frac{999999995}{1000000000}} = \frac{999999900}{999999995}$	1 1

	(iii) $(a)\ \frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt} = \frac{-5}{5^2+x^2} \times 20\Big]_{x=50}$ $= \frac{-100}{2525}$ or $\frac{-4}{101}\ rad/s$ **OR** **(b)** $\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt} \Rightarrow \frac{3}{101} = \frac{-5}{5^2+x^2}\Big]_{x=50} \times \frac{dx}{dt}$ $\Rightarrow \frac{3}{101} = \frac{-5}{2525} \times \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = -15\ m/s$ **Hence the speed is 15 m/s**	$1\frac{1}{2}$ $\frac{1}{2}$ $1\frac{1}{2}$ $\frac{1}{2}$
37.	According to recent research, air turbulence has increased in various regions around the world due to climate change. Turbulence makes flights bumpy and often delays the flights. Assume that, an airplane observes severe turbulence, moderate turbulence or light turbulence with equal probabilities. Further, the chance of an airplane reaching late to the destination are 55%, 37% and 17% due to severe, moderate and light turbulence respectively.  On the basis of the above information, answer the following questions : (i) Find the probability that an airplane reached its destination late. 2 (ii) If the airplane reached its destination late, find the probability that it was due to moderate turbulence. 2	
Sol.	**(i) Let A denote the event of airplane reaching its destination late** E_1 = **severe turbulence** E_2 = **moderate turbulence** E_3 = **light turbulence** **P(A)** = P $(E_1)\ P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3)$ $= \frac{1}{3} \times \frac{55}{100} + \frac{1}{3} \times \frac{37}{100} + \frac{1}{3} \times \frac{17}{100}$ $= \frac{1}{3}\left(\frac{109}{100}\right) = \frac{109}{300}$	$\left.\begin{matrix}\\\\\\\end{matrix}\right\} \frac{1}{2}$ 1 $\frac{1}{2}$

	(ii) $P(E_2\|A) = \frac{P(E_2)P(A\|E_2)}{P(A)}$ $= \frac{\frac{1}{3}\times\frac{37}{100}}{\frac{109}{300}}$ $= \frac{37}{109}$	$1\frac{1}{2}$ $\frac{1}{2}$
38.	If a function $f : X \to Y$ defined as $f(x) = y$ is one-one and onto, then we can define a unique function $g : Y \to X$ such that $g(y) = x$, where $x \in X$ and $y = f(x)$, $y \in Y$. Function g is called the inverse of function f. The domain of sine function is R and function sine : $R \to R$ is neither one-one nor onto. The following graph shows the sine function.  $y = \sin x$ Let sine function be defined from set A to $[-1, 1]$ such that inverse of sine function exists, i.e., $\sin^{-1} x$ is defined from $[-1, 1]$ to A. On the basis of the above information, answer the following questions : (i) If A is the interval other than principal value branch, give an example of one such interval. *1* (ii) If $\sin^{-1}(x)$ is defined from $[-1, 1]$ to its principal value branch, find the value of $\sin^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}(1)$. *1* (iii) (a) Draw the graph of $\sin^{-1} x$ from $[-1, 1]$ to its principal value branch. *2* **OR** (iii) (b) Find the domain and range of $f(x) = 2\sin^{-1}(1 - x)$. *2*	
Sol.	**(i)** $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ **or any other interval corresponding to the domain [-1,1]** **(ii)** $\sin^{-1}\left(\frac{-1}{2}\right) - \sin^{-1}(1)$ $= \frac{-\pi}{6} - \frac{\pi}{2}$ $= \frac{-4\pi}{6}$ or $\frac{-2\pi}{3}$	**1** **1**

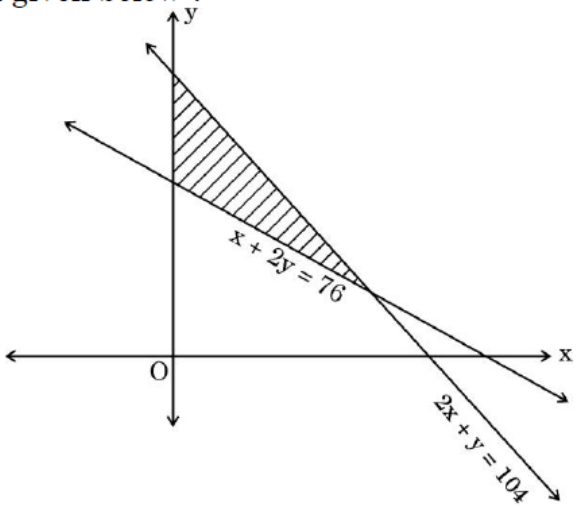
	(iii) (a)  	Correct graph - 2
	OR	
	(b) $f(x) = 2\,sin^{-1}(1-x)$	
	$-1 \leq 1-x \leq 1$	$\frac{1}{2}$
	$\Rightarrow -2 \leq -x \leq 0$	
	$\Rightarrow 0 \leq x \leq 2$	
	Domain $= [0, 2]$	$\frac{1}{2}$
	$\frac{-\pi}{2} \leq \sin^{-1}(1-x) \leq \frac{\pi}{2}$	$\frac{1}{2}$
	$-\pi \leq 2\,\sin^{-1}(1-x) \leq \pi$	
	So range $= [-\pi, \pi]$	$\frac{1}{2}$

Marking Scheme
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Senior School Certificate Examination, 2024
MATHEMATICS PAPER CODE 65/3/1

Q1	If $A = [a_{ij}]$ is an identity matrix, then which of the following is true ? (A) $a_{ij} = \begin{cases} 0, & \text{if } i = j \\ 1, & \text{if } i \neq j \end{cases}$ (B) $a_{ij} = 1, \forall\ i, j$ (C) $a_{ij} = 0, \forall\ i, j$ (D) $a_{ij} = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$	
Ans	(D) $a_{ij} = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$	1
Q2	Let R_+ denote the set of all non-negative real numbers. Then the function $f : R_+ \to R_+$ defined as $f(x) = x^2 + 1$ is : (A) one-one but not onto (B) onto but not one-one (C) both one-one and onto (D) neither one-one nor onto	
Ans	(A) one-one but not onto	1
Q3	Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a square matrix such that adj A = A. Then, $(a + b + c + d)$ is equal to : (A) 2a (B) 2b (C) 2c (D) 0	
Ans	(A) 2a	1
Q4	A function $f(x) = \lvert 1 - x + \lvert x \rvert \rvert$ is : (A) discontinuous at x = 1 only (B) discontinuous at x = 0 only (C) discontinuous at x = 0, 1 (D) continuous everywhere	
Ans	(D) continuous everywhere	1
Q5	If the sides of a square are decreasing at the rate of 1·5 cm/s, the rate of decrease of its perimeter is : (A) 1·5 cm/s (B) 6 cm/s (C) 3 cm/s (D) 2·25 cm/s	
Ans	(B) 6 cm/s	1

Q6	$\int_{-a}^{a} f(x)\,dx = 0$, if : (A) $f(-x) = f(x)$ (B) $f(-x) = -f(x)$ (C) $f(a-x) = f(x)$ (D) $f(a-x) = -f(x)$	
Ans	(B) $f(-x) = -f(x)$	1
Q7	$x \log x \frac{dy}{dx} + y = 2 \log x$ is an example of a : (A) variable separable differential equation. (B) homogeneous differential equation. (C) first order linear differential equation. (D) differential equation whose degree is not defined.	
Ans	(C) first order linear differential equation.	1
Q8	If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$, then $\vec{a}$ and $\vec{b}$ are : (A) collinear vectors which are not parallel (B) parallel vectors (C) perpendicular vectors (D) unit vectors	
Ans	(C) perpendicular vectors	1
Q9	If α, β and γ are the angles which a line makes with positive directions of x, y and z axes respectively, then which of the following is ***not*** true ? (A) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ (B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ (C) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$ (D) $\cos\alpha + \cos\beta + \cos\gamma = 1$	
Ans	(D) $\cos\alpha + \cos\beta + \cos\gamma = 1$	1
Q10	The restrictions imposed on decision variables involved in an objective function of a linear programming problem are called : (A) feasible solutions (B) constraints (C) optimal solutions (D) infeasible solutions	
Ans	(B) constraints	1
Q11	Let E and F be two events such that P(E) = 0·1, P(F) = 0·3, P(E $\cup$ F) = 0·4, then P(F | E) is : (A) 0·6 (B) 0·4 (C) 0·5 (D) 0	

Ans	(D) 0	1
Q12	If A and B are two skew symmetric matrices, then (AB + BA) is : (A) a skew symmetric matrix (B) a symmetric matrix (C) a null matrix (D) an identity matrix	
Ans	(B) a symmetric matrix	1
Q13	If $\begin{vmatrix} 1 & 3 & 1 \\ k & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \pm 6$, then the value of k is : (A) 2 (B) -2 (C) ± 2 (D) ∓ 2	
Ans	(D) ∓ 2	1
Q14	The derivative of 2^x w.r.t. 3^x is : (A) $\left(\frac{3}{2}\right)^x \frac{\log 2}{\log 3}$ (B) $\left(\frac{2}{3}\right)^x \frac{\log 3}{\log 2}$ (C) $\left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$ (D) $\left(\frac{3}{2}\right)^x \frac{\log 3}{\log 2}$	
Ans	(C) $\left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$	1
Q15	If $\lvert \vec{a} \rvert = 2$ and $-3 \le k \le 2$, then $\lvert k\vec{a} \rvert \in$: (A) $[-6, 4]$ (B) $[0, 4]$ (C) $[4, 6]$ (D) $[0, 6]$	
Ans	(D) $[0, 6]$	1
Q16	If a line makes an angle of $\frac{\pi}{4}$ with the positive directions of both x-axis and z-axis, then the angle which it makes with the positive direction of y-axis is : (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π	
Ans	(C) $\frac{\pi}{2}$	1

Q17	Of the following, which group of constraints represents the feasible region given below ?  (A) $x + 2y \leq 76, 2x + y \geq 104, x, y \geq 0$ (B) $x + 2y \leq 76, 2x + y \leq 104, x, y \geq 0$ (C) $x + 2y \geq 76, 2x + y \leq 104, x, y \geq 0$ (D) $x + 2y \geq 76, 2x + y \geq 104, x, y \geq 0$	
Ans	(C) $x + 2y \geq 76, 2x + y \leq 104, x, y \geq 0$	1
Q18	If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$, then A^{-1} is : (A) $\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$ (B) $30\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$ (C) $\frac{1}{30}\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ (D) $\frac{1}{30}\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$	
Ans	(A) $\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$	1

*Questions number **19** and **20** are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Q19	*Assertion (A) :* Every scalar matrix is a diagonal matrix. *Reason (R) :* In a diagonal matrix, all the diagonal elements are 0.	
Ans	(C) Assertion (A) is true, but Reason (R) is false.	1
Q20	*Assertion (A) :* Projection of $\vec{a}$ on $\vec{b}$ is same as projection of $\vec{b}$ on $\vec{a}$. *Reason (R) :* Angle between $\vec{a}$ and $\vec{b}$ is same as angle between $\vec{b}$ and $\vec{a}$ numerically.	
Ans	(D) Assertion (A) is false, but Reason (R) is true.	1

SECTION B

Questions no. 21 to 25 are very short answer (VSA) type questions, carrying 2 marks each.

Q21	Evaluate : $\sec^2\left(\tan^{-1}\frac{1}{2}\right) + \text{cosec}^2\left(\cot^{-1}\frac{1}{3}\right)$	
Ans	$\sec^2\left(\tan^{-1}\frac{1}{2}\right)+\text{cosec}^2\left(\cot^{-1}\frac{1}{3}\right)$ $=\left[1+\tan^2\left(\tan^{-1}\frac{1}{2}\right)\right]+\left[1+\cot^2\left(\cot^{-1}\frac{1}{3}\right)\right]$ $=\left[1+\left(\frac{1}{2}\right)^2\right]+\left[1+\left(\frac{1}{3}\right)^2\right]$ $=\frac{85}{36}$	1 ½ ½
Q22(a)	If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$	

Ans	$x = e^{\frac{x}{y}} \Rightarrow \log x = \frac{x}{y} \Rightarrow y = \frac{x}{\log x}$ $\Rightarrow \frac{dy}{dx} = \frac{(\log x)(1) - x\left(\frac{1}{x}\right)}{(\log x)^2} = \frac{\log x - 1}{(\log x)^2}$	1 1
	OR	
Q22(b)	Check the differentiability of f(x) = $\begin{cases} x^2 + 1, & 0 \le x < 1 \\ 3 - x, & 1 \le x \le 2 \end{cases}$ at x = 1.	
Ans	LHD at $x = 1$ $= \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \to 0} \frac{\left[(1-h)^2 + 1\right] - 2}{-h} = 2$ RHD at $x = 1$ $= \lim_{h \to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0} \frac{\left[3 - (1+h)\right] - 2}{h} = -1$ as LHD $\neq$ RHD, so $f(x)$ is not differentiable at $x = 1$	1 ½ ½
Q23(a)	Evaluate : $\int_0^{\pi/2} \sin 2x \cos 3x \, dx$	
Ans	$I = \int_0^{\frac{\pi}{2}} \sin 2x \cos 3x \, dx$ $= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\sin 5x - \sin x) \, dx$ $= \frac{1}{2}\left[-\frac{1}{5}\cos 5x + \cos x\right]_0^{\frac{\pi}{2}}$ $= -\frac{2}{5}$	1 ½ ½
	OR	
Q23(b)	Given $\frac{d}{dx} F(x) = \frac{1}{\sqrt{2x - x^2}}$ and F(1) = 0, find F(x).	

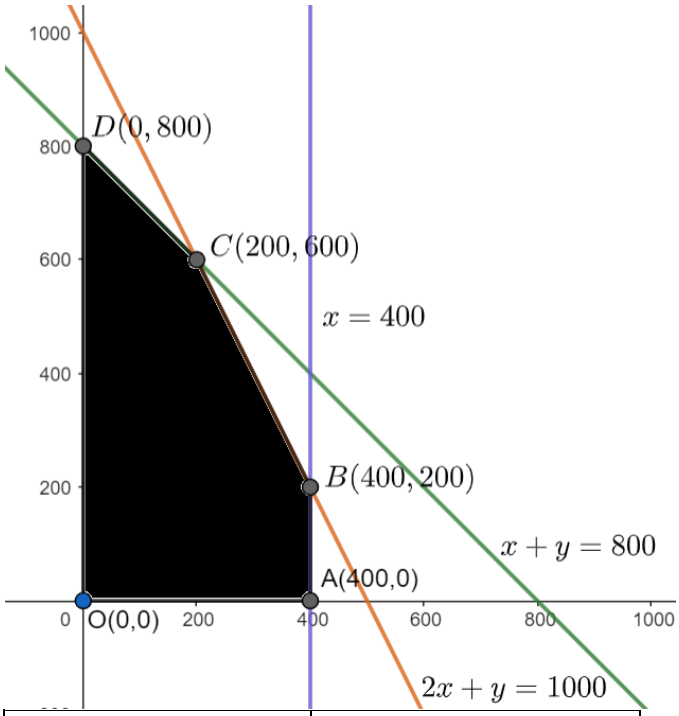
Ans	$F(x)=\int \frac{1}{\sqrt{2x-x^2}}dx$ $=\int \frac{1}{\sqrt{1-(x-1)^2}}dx$ $=\sin^{-1}(x-1)+c$ when $x=1, y=0$ gives $c=0$ $\therefore F(x)=\sin^{-1}(x-1)$	½ ½ ½ ½
Q24	Find the position vector of point C which divides the line segment joining points A and B having position vectors $\hat{i}+2\hat{j}-\hat{k}$ and $-\hat{i}+\hat{j}+\hat{k}$ respectively in the ratio 4 : 1 externally. Further, find $\lvert\vec{AB}\rvert : \lvert\vec{BC}\rvert$.	
Ans	Position vector of $C=\vec{r}=\frac{4\vec{b}-\vec{a}}{3}$ i.e. $\vec{r}=\frac{1}{3}\left(-5\hat{i}+2\hat{j}+5\hat{k}\right)$ Now, $\vec{AB}=-2\hat{i}-\hat{j}+2\hat{k}\Rightarrow\left\lvert\vec{AB}\right\rvert=3$ $\vec{BC}=-\frac{1}{3}\left(2\hat{i}+\hat{j}-2\hat{k}\right)\Rightarrow\left\lvert\vec{BC}\right\rvert=1$ $\left\lvert\vec{AB}\right\rvert:\left\lvert\vec{BC}\right\rvert=3:1$	1 1
Q25	Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors. Prove that $\lvert\vec{a}\times\vec{b}\rvert \le \lvert\vec{a}\rvert\lvert\vec{b}\rvert$. State the condition under which equality holds, i.e., $\lvert\vec{a}\times\vec{b}\rvert = \lvert\vec{a}\rvert\lvert\vec{b}\rvert$.	
Ans	$\lvert\vec{a}\times\vec{b}\rvert=\lvert\vec{a}\rvert\lvert\vec{b}\rvert\lvert\sin\theta\rvert$ As, $0\le\lvert\sin\theta\rvert\le 1$ $\Rightarrow\lvert\vec{a}\rvert\lvert\vec{b}\rvert\lvert\sin\theta\rvert\le\lvert\vec{a}\rvert\lvert\vec{b}\rvert$ $\Rightarrow\lvert\vec{a}\times\vec{b}\rvert\le\lvert\vec{a}\rvert\lvert\vec{b}\rvert$ For equality, $\sin\theta=1\Rightarrow\theta=\frac{\pi}{2}\Rightarrow\vec{a}$ is perpendicular to $\vec{b}$.	½ ½ ½ ½
SECTION C Questions no. 26 to 31 are short answer (SA) type questions, carrying 3 marks each.		

Q26(a)	If x cos (p + y) + cos p sin (p + y) = 0, prove that $\cos p \frac{dy}{dx} = -\cos^2 (p + y)$, where p is a constant.	
Ans	$x\cos(p+y)+\cos p\sin(p+y)=0$ $\Rightarrow x=\frac{-\cos p\sin(p+y)}{\cos(p+y)} \Rightarrow x=-\cos p.\tan(p+y)$ $\Rightarrow \frac{dx}{dy}=-\cos p.\sec^2(p+y)$ $\Rightarrow \frac{dy}{dx}=\frac{-1}{\cos p.\sec^2(p+y)}$ $\Rightarrow \cos p\frac{dy}{dx}=-\cos^2(p+y)$	1 1 ½ ½
	OR	
Q26(b)	Find the value of a and b so that function f defined as : $f(x)=\begin{cases}\frac{x-2}{\lvert x-2\rvert}+a, & \text{if } x<2\\ a+b, & \text{if } x=2\\ \frac{x-2}{\lvert x-2\rvert}+b, & \text{if } x>2\end{cases}$ is a continuous function.	
Ans	$f(x)=\begin{cases}\frac{x-2}{-(x-2)}+a & ;x<2\\ a+b & ;x=2\\ \frac{x-2}{(x-2)}+b & ;x>2\end{cases} \Rightarrow f(x)=\begin{cases}-1+a & ;x<2\\ a+b & ;x=2\\ 1+b & ;x>2\end{cases}$ $\lim_{x\to 2^-} f(x)=-1+a, \lim_{x\to 2^+} f(x)=1+b$ and $f(2)=a+b$ as f is continous at $x=2 \therefore -1+a=1+b=a+b$ $\Rightarrow a=1, b=-1$	1 1 ½+½
Q27(a)	Find the intervals in which the function f(x) = $\frac{\log x}{x}$ is strictly increasing or strictly decreasing.	
Ans	$f(x)=\frac{\log x}{x} \Rightarrow f'(x)=\frac{1-\log x}{x^2} ; x>0$ for strictly increasing/decreasing, put $f'(x)=0 \Rightarrow x=e$ for strictly increasing, $x\in(0,e)$ and for strictly decreasing $x\in(e,\infty)$	1 1 ½+½

Q27(b)	Find the absolute maximum and absolute minimum values of the function f given by $f(x) = \frac{x}{2} + \frac{2}{x}$, on the interval [1, 2].	
Ans	$f(x) = \frac{x}{2} + \frac{2}{x}$; $x \in [1,2]$ $\Rightarrow f'(x) = \frac{1}{2} - \frac{2}{x^2}$ for absolute maximum / minimum, put $f'(x) = 0$ $\Rightarrow x^2 = 4 \Rightarrow x = 2$ Now, $f(1) = \frac{5}{2}$ and $f(2) = 2$ $\therefore$ absolute maximum value $= \frac{5}{2}$ and absolute minimum value $= 2$	1 ½ ½+½ ½
Q28	Find : $\int \frac{x^2+1}{(x^2+2)(x^2+4)}\,dx$	
Ans	$I = \int \frac{x^2+1}{(x^2+2)(x^2+4)}\,dx$ Let $x^2 = y$, then $\frac{x^2+1}{(x^2+2)(x^2+4)} = \frac{y+1}{(y+2)(y+4)}$ Let $\frac{y+1}{(y+2)(y+4)} = \frac{A}{y+2} + \frac{B}{y+4}$ this gives $A = -\frac{1}{2}, B = \frac{3}{2}$ $\therefore I = -\frac{1}{2}\int \frac{1}{x^2+2}\,dx + \frac{3}{2}\int \frac{1}{x^2+4}\,dx$ $\Rightarrow I = -\frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{3}{4}\tan^{-1}\left(\frac{x}{2}\right) + c$	1 1 1
Q29(a)	Find : $\int \frac{2+\sin 2x}{1+\cos 2x} e^x\,dx$	

Ans	$I=\int \frac{2+\sin 2x}{1+\cos 2x}e^x\,dx$ $=\int \frac{2+2\sin x\cos x}{2\cos^2 x}e^x\,dx$ $=\int \left(\sec^2 x+\tan x\right)e^x dx$ $=e^x.\tan x+c$	1 1 1
OR		
Q29(b)	Evaluate : $\int_0^{\pi/4} \frac{1}{\sin x+\cos x}\,dx$	
Ans	$I=\int_0^{\frac{\pi}{4}} \frac{1}{\sin x+\cos x}dx$ $=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}} \frac{1}{\cos\frac{\pi}{4}\sin x+\sin\frac{\pi}{4}\cos x}dx$ $=\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}} \frac{1}{\sin\left(x+\frac{\pi}{4}\right)}dx==\frac{1}{\sqrt{2}}\int_0^{\frac{\pi}{4}} \operatorname{cosec}\left(x+\frac{\pi}{4}\right)dx$ $=\frac{1}{\sqrt{2}}\left[\log\left|\operatorname{cosec}\left(x+\frac{\pi}{4}\right)-\cot\left(x+\frac{\pi}{4}\right)\right|\right]_0^{\frac{\pi}{4}}$ $=\frac{1}{\sqrt{2}}\log\left(\sqrt{2}+1\right)\text{or}-\frac{1}{\sqrt{2}}\log\left(\sqrt{2}-1\right)$	1 1 1
Q30	Solve the following linear programming problem graphically : Maximise z = 4x + 3y, subject to the constraints $x + y \le 800$ $2x + y \le 1000$ $x \le 400$ $x, y \ge 0.$	

Ans



For correct graph
1½

Corner Point	Value of $z=4x+3y$
$O(0,0)$	0
$A(400,0)$	1600
$B(400,200)$	2200
$C(200,600)$	2600
$D(0,800)$	2400

For correct table
1

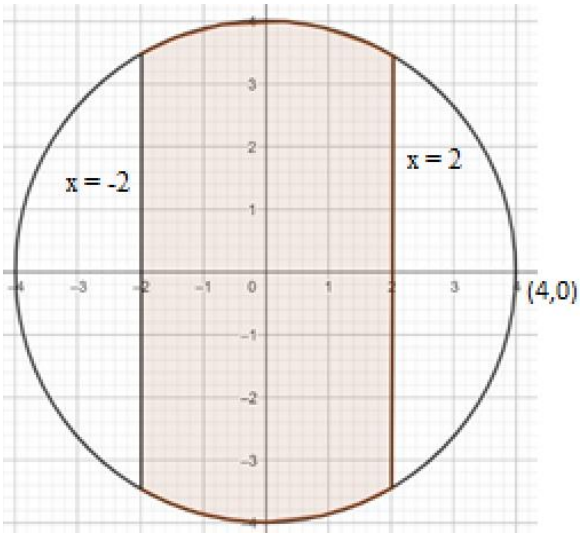
$z_{max}=2600$ when $x=200, y=600$

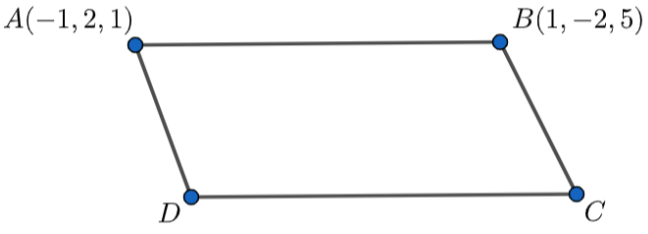
½

Q31

The chances of P, Q and R getting selected as CEO of a company are in the ratio 4 : 1 : 2 respectively. The probabilities for the company to increase its profits from the previous year under the new CEO, P, Q or R are 0·3, 0·8 and 0·5 respectively. If the company increased the profits from the previous year, find the probability that it is due to the appointment of R as CEO.

Ans	Let E_1 : P is appointed as CEO, E_2 : Q is appointed as CEO, E_3 : R is appointed as CEO A : company increase profits from previous year	½
	here, $P(E_1)=\frac{4}{7}, P(E_3)=\frac{1}{7}, P(E_1)=\frac{2}{7}$ $P(A\mid E_1)=0.3, P(A\mid E_2)=0.8, P(A\mid E_3)=0.5$	1
	$P(E_3\mid A)=\frac{P(E_3)P(A\mid E_3)}{P(E_1)P(A\mid E_1)+P(E_2)P(A\mid E_2)+P(E_3)P(A\mid E_3)}$ $=\frac{\frac{2}{7}\times 0.5}{\frac{4}{7}\times 0.3+\frac{1}{7}\times 0.8+\frac{2}{7}\times 0.5}$	1
	$=\frac{1}{3}$	½
SECTION D Questions no. 32 to 35 are long answer (LA) type questions carrying 5 marks each.		
Q32	A relation R on set A = {– 4, – 3, – 2, – 1, 0, 1, 2, 3, 4} be defined as R = {(x, y) : x + y is an integer divisible by 2}. Show that R is an equivalence relation. Also, write the equivalence class [2].	
Ans	For reflexive: clearly $x+x$ i.e. $2x$ is integer divisible by 2. $\Rightarrow (x,x)\in R \Rightarrow R$ is reflexive.	1
	For symmetric : $(x,y)\in R \Rightarrow x+y$ is integer divisible by 2. $\Rightarrow y+x$ is integer divisible by $2 \Rightarrow (y,x)\in R$	1
	For transitive : $(x,y)\in R \Rightarrow x+y$ is integer divisible by 2. and $(y,z)\in R \Rightarrow y+z$ is integer divisible by 2. so, $(x+z)+2y$ is integer divisible by 2. $\Rightarrow x+z$ is integer divisible by $2 \Rightarrow (x,z)\in R$	2
	Equivalence class $[2]=\{-4,-2,0,2,4\}$	1
Q33(a)	It is given that function f(x) = $x^4 - 62x^2 + ax + 9$ attains local maximum value at x = 1. Find the value of 'a', hence obtain all other points where the given function f(x) attains local maximum or local minimum values.	

Ans	$f(x)=x^4-62x^2+ax+9 \Rightarrow f'(x)=4x^3-124x+a$	½
	as at $x=1$, f attains local maximum value, $f'(1)=0 \Rightarrow a=120$	1
	now, $f'(x)=4x^3-124x+120=4(x-1)(x^2+x-30)=4(x-1)(x-5)(x+6)$	1
	Critical points are $x=-6,1,\ 5$	1
	$f''(x)=12x^2-124$	
	$f''(-6)>0,\ f''(1)<0\ ,\ f''(5)>0$	½
	so f attains local maximum value at $x=1$ and local minimum value at $x=-6,\ 5$	1
	OR	
Q33(b)	The perimeter of a rectangular metallic sheet is 300 cm. It is rolled along one of its sides to form a cylinder. Find the dimensions of the rectangular sheet so that volume of cylinder so formed is maximum.	
Ans	Let length of rectangle be x cm and breadth be $(150-x)$ cm.	
	Let r be the radius of cylinder $\Rightarrow 2\pi r=x \Rightarrow r=\frac{x}{2\pi}$	1
	$V=\pi r^2 h=\pi\left(\frac{x^2}{4\pi^2}\right)(150-x)=\frac{75x^2}{2\pi}-\frac{x^3}{4\pi}$	1
	$\frac{dV}{dx}=\frac{150x}{2\pi}-\frac{3x^2}{4\pi}$	1
	$\frac{dV}{dx}=0 \Rightarrow x=100$ cm	1
	$\left.\frac{d^2V}{dx^2}\right\|_{x=100\text{ cm}}=-\frac{75}{\pi}<0 \Rightarrow V$ is maximum when $x=100$ cm.	½
	Length of rectangle is 100 cm and breadth of rectangle is 50 cm.	½
Q34	Using integration, find the area of the region enclosed between the circle $x^2 + y^2 = 16$ and the lines $x = -2$ and $x = 2$.	
Ans		For correct figure 1 mark

	Required area $= 4\int_0^2 \sqrt{16-x^2}\,dx$	1
	$= 4\left[\frac{x}{2}\sqrt{16-x^2} + 8\sin^{-1}\left(\frac{x}{4}\right)\right]_0^2$	2
	$= 8\sqrt{3} + \frac{16\pi}{3}$	1
Q35(a)	Find the equation of the line passing through the point of intersection of the lines $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ and $\frac{x-1}{0} = \frac{y}{-3} = \frac{z-7}{2}$ and perpendicular to these given lines.	
Ans	$l_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \lambda$; $l_2 : \frac{x-1}{0} = \frac{y}{-3} = \frac{z-7}{2} = \mu$	
	any point on l_1 is $(\lambda, 2\lambda+1, 3\lambda+2)$ & any point on l_2 is $(1, -3\mu, 2\mu+7)$	1
	If l_1 and l_2 intersect,	
	$\lambda = 1, 2\lambda+1 = -3\mu$ and $3\lambda+2 = 2\mu+7 \Rightarrow \lambda = 1$ and $\mu = -1$	1
	Point of intersection of l_1 and l_2 is $(1,3,5)$.	1
	Let d.r.'s of required line be $< a, b, c >$. Then,	
	$a + 2b + 3c = 0$ and $-3b + 2c = 0 \Rightarrow \frac{a}{13} = \frac{b}{-2} = \frac{c}{-3}$	1
	Required equation of line is $\frac{x-1}{13} = \frac{y-3}{-2} = \frac{z-5}{-3}$	1
	OR	
Q35(b)	Two vertices of the parallelogram ABCD are given as A(−1, 2, 1) and B(1, −2, 5). If the equation of the line passing through C and D is $\frac{x-4}{1} = \frac{y+7}{-2} = \frac{z-8}{2}$, then find the distance between sides AB and CD. Hence, find the area of parallelogram ABCD.	
Ans		
	d.r's of CD are $< 1, -2, 2 >$	
	$\therefore$ d.r's of AB are $< 1, -2, 2 >$	½

	$\therefore$ Equation of AB is $\frac{x+1}{1} = \frac{y-2}{-2} = \frac{z-1}{2}$	½
	$\therefore$ Equation of CD is $\frac{x-4}{1} = \frac{y+7}{-2} = \frac{z-8}{2}$	
	Let $\vec{a}_1 = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{a}_2 = 4\hat{i} - 7\hat{j} + 8\hat{k}$ & $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ Now, $\vec{a}_2 - \vec{a}_1 = 5\hat{i} - 9\hat{j} + 7\hat{k}$	1
	$(\vec{a}_2 - \vec{a}_1) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -9 & 7 \\ 1 & -2 & 2 \end{vmatrix} = -4\hat{i} - 3\hat{j} - \hat{k}$	1
	Distance between AB and CD is given by $d = \frac{\lvert(\vec{a}_2 - \vec{a}_1) \times \vec{b}\rvert}{\lvert\vec{b}\rvert}$	
	$d = \frac{\sqrt{16+9+1}}{\sqrt{1+4+4}} = \frac{\sqrt{26}}{3}$	½
	$CD = \sqrt{2^2 + (-4)^2 + (4)^2} = 6$	½
	Area of parallelogram ABCD $= b \times h = 6 \times \frac{\sqrt{26}}{3} = 2\sqrt{26}$	1

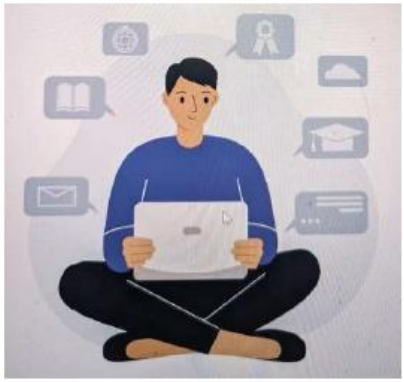
SECTION E

Questions no. 36 to 38 are case study based questions carrying 4 marks each.

Q36

Self-study helps students to build confidence in learning. It boosts the self-esteem of the learners. Recent surveys suggested that close to 50% learners were self-taught using internet resources and upskilled themselves.





A student may spend 1 hour to 6 hours in a day in upskilling self. The probability distribution of the number of hours spent by a student is given below :

$$P(X = x) = \begin{cases} kx^2, & \text{for } x = 1, 2, 3 \\ 2kx, & \text{for } x = 4, 5, 6 \\ 0, & \text{otherwise} \end{cases}$$

where x denotes the number of hours.

Based on the above information, answer the following questions :

(i) Express the probability distribution given above in the form of a probability distribution table. *1*

(ii) Find the value of k. *1*

(iii) (a) Find the mean number of hours spent by the student. *2*

OR

(iii) (b) Find $P(1 < X < 6)$. *2*

Ans(i)

X	1	2	3	4	5	6
P(X)	k	4k	9k	8k	10k	12k

1

Ans(ii)

$k + 4k + 9k + 8k + 10k + 12k = 1$

$\Rightarrow k = \frac{1}{44}$

1

Ans (iii) (a)

$\text{Mean} = \sum x_i p_i = k + 8k + 27k + 32k + 50k + 72k$ — 1

$= 190k$

$= \frac{190}{44} \text{ or } \frac{95}{22}$ — 1

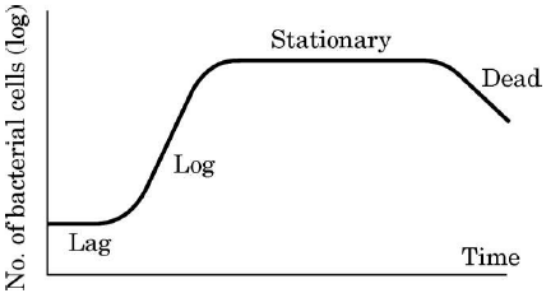
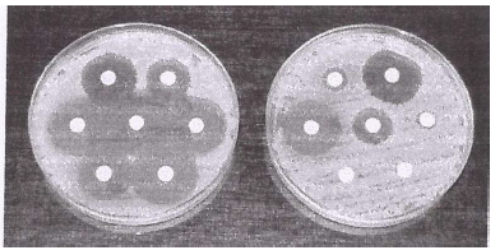
OR

Ans (iii)(b)

$P(1 < X < 6) = 4k + 9k + 8k + 10k$ — 1

$= 31k$

$= \frac{31}{44}$ — 1

Q37	A bacteria sample of certain number of bacteria is observed to grow exponentially in a given amount of time. Using exponential growth model, the rate of growth of this sample of bacteria is calculated.   The differential equation representing the growth of bacteria is given as : $\frac{dP}{dt} = kP$, where P is the population of bacteria at any time 't'. Based on the above information, answer the following questions : (i) Obtain the general solution of the given differential equation and express it as an exponential function of 't'. *2* (ii) If population of bacteria is 1000 at t = 0, and 2000 at t = 1, find the value of k. *2*	
Ans(i)	$\frac{dP}{dt} = kP \Rightarrow \int \frac{dP}{P} = \int k\,dt$ $\Rightarrow \log P = kt + C$ or $P = e^{kt+C}$	1 1
Ans(ii)	$\log P = kt + C$ when $t = 0, P = 1000 \Rightarrow C = \log 1000$ when $t = 1, P = 2000 \Rightarrow \log 2000 = k + \log 1000$ $\Rightarrow k = \log 2$	½ ½ 1

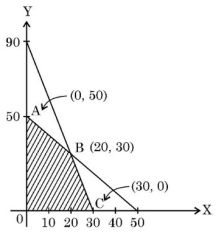
Q38	A scholarship is a sum of money provided to a student to help him or her pay for education. Some students are granted scholarships based on their academic achievements, while others are rewarded based on their financial needs.   Every year a school offers scholarships to girl children and meritorious achievers based on certain criteria. In the session 2022 – 23, the school offered monthly scholarship of ₹ 3,000 each to some girl students and ₹ 4,000 each to meritorious achievers in academics as well as sports. In all, 50 students were given the scholarships and monthly expenditure incurred by the school on scholarships was ₹ 1,80,000. Based on the above information, answer the following questions : (i) Express the given information algebraically using matrices. *1* (ii) Check whether the system of matrix equations so obtained is consistent or not. *1* (iii) (a) Find the number of scholarships of each kind given by the school, using matrices. *2* **OR** (iii) (b) Had the amount of scholarship given to each girl child and meritorious student been interchanged, what would be the monthly expenditure incurred by the school ? *2*	
Ans(i)	Let No. of girl child scholarships = x No. of meritorious achievers = y $x + y = 50$ $3000x + 4000y = 180000$ or $3x + 4y = 180$ $\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 50 \\ 180 \end{bmatrix}$	1
Ans(ii)	$\begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} = 1 \neq 0$	

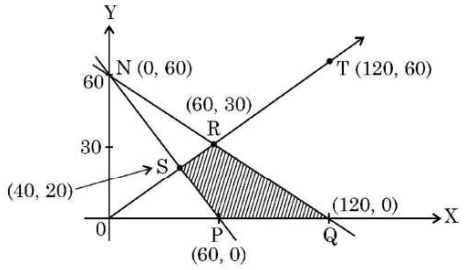
	$\therefore$ system is consistent.	1
Ans (iii)(a)	Let $A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $B = \begin{bmatrix} 50 \\ 180 \end{bmatrix}$ $AX = B \Rightarrow X = A^{-1}B$ $X = \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 50 \\ 180 \end{bmatrix} = \begin{bmatrix} 20 \\ 30 \end{bmatrix}$ $\Rightarrow x = 20, y = 30$	½ 1 ½
OR		
Ans (iii)(b)	Required expenditure = ₹ [30(3000) + 20(4000)] = ₹ 1,70,000	1 1

Marking Scheme
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Senior School Certificate Examination, 2024
MATHEMATICS PAPER CODE 65/4/1

1.	If $\begin{bmatrix} a & c & 0 \\ b & d & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is a scalar matrix, then the value of $a+2b+3c+4d$ is $(A)\ 0$ $(B)\ 5$ $(C)\ 10$ $(D)\ 25$	
Ans:	**(D) 25**	**1**
2.	Given that $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$, matrix A is: $(A)\ 7\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ $(B)\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ $(C)\frac{1}{7}\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ $(D)\frac{1}{49}\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$	
Ans:	$(B)\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$	**1**
3.	If $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$, then the value of $I - A + A^2 - A^3 + \ldots$ is $(A)\begin{bmatrix} -1 & -1 \\ 4 & 3 \end{bmatrix}$ $(B)\begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix}$ $(C)\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $(D)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	
Ans:	$(A)\begin{bmatrix} -1 & -1 \\ 4 & 3 \end{bmatrix}$	**1**
4.	If $A = \begin{bmatrix} -2 & 0 & 0 \\ 1 & 2 & 3 \\ 5 & 1 & -1 \end{bmatrix}$, then the value of $\lvert A\ (\text{adj. } A) \rvert$ is : $(A)\ 100\ I$ $(B)\ 10\ I$ $(C)10$ $(D)1000$	
Ans:	**(D) 1000**	**1**
5.	Given that $\begin{bmatrix} 1 & x \end{bmatrix}\begin{bmatrix} 4 & 0 \\ -2 & 0 \end{bmatrix} = 0$, then value of x is : $(A)\ -4$ $(B)\ -2$ $(C)\ 2$ $(D)\ 4$	
Ans:	**(C) 2**	**1**
6.	Derivative of e^{2x} with respect to e^{x}, is : $(A)\ e^{x}$ $(B)\ 2e^{x}$ $(C)\ 2e^{2x}$ $(D)\ 2e^{3x}$	
Ans:	$(B)\ 2e^{x}$	**1**

7.	For what value of k, the function given below is continuous at $x = 0$? $f(x) = \begin{cases} \frac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ $(A)\ 0 \qquad (B)\ \frac{1}{4} \qquad (C)\ 1 \qquad (D)\ 4$	
Ans:	$(B)\ \frac{1}{4}$	**1**
8.	The value of $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ is : $(A)\ \frac{\pi}{6} \qquad (B)\ \frac{\pi}{4} \qquad (C)\ \frac{\pi}{2} \qquad (D)\ \frac{\pi}{18}$	
Ans:	$(C)\ \frac{\pi}{2}$	**1**
9.	The general solution of the differential equation $x\ dy + y\ dx = 0$ is : (A) $xy = c$ (B) $x + y = c$ (C) $x^2 + y^2 = c^2$ (D) $\log y = \log x + c$	
Ans:	$(A)\ xy = c$	**1**
10.	The integrating factor of the differential equation $(x + 2y^2)\frac{dy}{dx} = y\ \ (y > 0)$ is : $(A)\ \frac{1}{x} \qquad (B)\ x \qquad (C)\ y \qquad (D)\ \frac{1}{y}$	
Ans:	**(D)** $\frac{1}{y}$	**1**
11.	If $\vec{a}$ and $\vec{b}$ are two vectors such that $\lvert\vec{a}\rvert = 1, \lvert\vec{b}\rvert = 2$ and $\vec{a}.\vec{b} = \sqrt{3}$, then the angle between $2\vec{a}$ and $-\vec{b}$ is: $(A)\ \frac{\pi}{6} \qquad (B)\ \frac{\pi}{3} \qquad (C)\ \frac{5\pi}{6} \qquad (D)\ \frac{11\pi}{6}$	
Ans:	**(C)** $\frac{5\pi}{6}$	**1**
12.	The vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}, \vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -3\hat{i} + 4\hat{j} + 4\hat{k}$ represents the sides of (A) an equilaterl triangle (B) an obtuse-angled triangle (C) an isosceles triangle (D) a right-angled triangle	
Ans:	**(D) a right-angled triangle**	**1**

13.	Let $\vec{a}$ be any vector such that $\|\vec{a}\| = a$. The value of $\|\vec{a} \times \hat{i}\|^2 + \|\vec{a} \times \hat{j}\|^2 + \|\vec{a} \times \hat{k}\|^2$ is: $(A)\ a^2$ $(B)\ 2a^2$ $(C)\ 3a^2$ $(D)\ 0$	
Ans:	**(B)** $2a^2$	**1**
14.	The vector equation of a line passing through the point (1, –1, 0) and parallel to Y-axis is : (A) $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} - \hat{j})$ (B) $\vec{r} = \hat{i} - \hat{j} + \lambda \hat{j}$ (C) $\vec{r} = \hat{i} - \hat{j} + \lambda \hat{k}$ (D) $\vec{r} = \lambda \hat{j}$	
Ans:	**(B)** $\vec{r} = \hat{i} - \hat{j} + \lambda \hat{j}$	**1**
15.	The lines $\frac{1-x}{2} = \frac{y-1}{3} = \frac{z}{1}$ and $\frac{2x-3}{2p} = \frac{y}{-1} = \frac{z-4}{7}$ are perpendicular to each other for p equal to : (A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 2 (D) 3	
Ans:	**(C) 2**	**1**
16.	The maximum value of Z = 4x + y for a L.P.P. whose feasible region is given below is  $(A)\ 50$ $(B)\ 110$ $(C)\ 120$ $(D)\ 170$	
Ans:	**(C) 120**	**1**
17.	The probability distribution of a random variable X is: **X** 0 1 2 3 4 **P(X)** 0.1 k 2k k 0.1 where k is some unknown constant. The probability that the random variable X takes the value 2 is: $(A)\ \frac{1}{5}$ $(B)\ \frac{2}{5}$ $(C)\ \frac{4}{5}$ $(D)\ 1$	

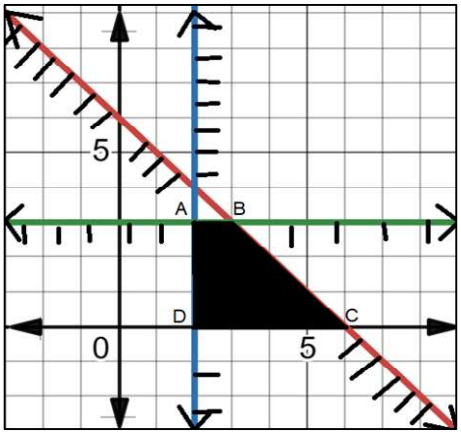
Ans:	**(B)** $\frac{2}{5}$	**1**
18.	The function $f(x) = kx - \sin x$ is strictly increasing for (A) $k > 1$ (B) $k < 1$ (C) $k > -1$ (D) $k < -1$	
Ans:	**(A) k > 1**	**1**
	ASSERTION-REASON BASED QUSTIONS Questions No. 19 & 20, are Assertion (A) and Reason (R) based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct nswer from the codes (A), (B), (C) and (D) as given below: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	
19.	**Assertion (A) :** The relation $R = \{(x, y) : (x + y)$ is a prime number and $x, y \in N\}$ is not a reflexive relation. **Reason (R) :** The number '2n' is composite for all natural numbers n.	
Ans:	**(C) Assertion (A) is true, but Reason (R) is false.**	**1**
20.	**Assertion (A) :** The corner points of the bounded feasible region of a L.P.P. are shown below. The maximum value of $Z = x + 2y$ occurs at infinite points.  **Reason (R) :** The optimal solution of a LPP having bounded feasible region must occur at corner points.	
Ans:	**(B) Both A and R are true but R is not the correct explanation of A.**	**1**
	SECTION B **In this section there are 5 very short answer type questions of 2 marks each.**	
21(a).	Express $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$, where $\frac{-\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.	

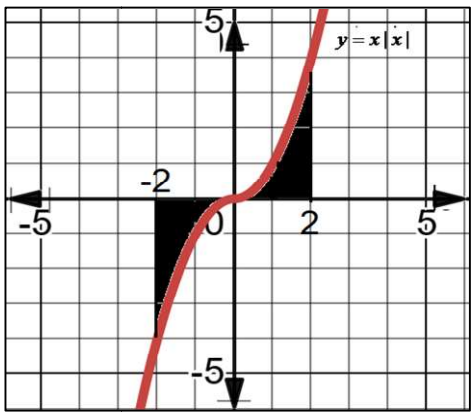
Sol.	$y = \tan^{-1}\left[\frac{\cos x}{1-\sin x}\right] = \tan^{-1}\left[\frac{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}{(\cos\frac{x}{2}-\sin\frac{x}{2})^2}\right]$ $y = \tan^{-1}\left[\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}\right] = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right] = \left(\frac{\pi}{4}+\frac{x}{2}\right)$	½ 1 ½
	OR	
21(b).	Find the principal value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$.	
Sol.	$\tan^{-1}(1)+\left[\pi-\cos^{-1}(\frac{1}{2})\right]-\sin^{-1}(\frac{1}{\sqrt{2}}) = \frac{\pi}{4}+\left(\pi-\frac{\pi}{3}\right)-\frac{\pi}{4}$ $= \frac{2\pi}{3}$	1 ½ ½
22(a).	If $y = \cos^3(\sec^2 2t)$, find $\frac{dy}{dt}$.	
Sol.	$y = \cos^3(\sec^2 2t)$ $\Rightarrow \frac{dy}{dt} = 3\cos^2(\sec^2 2t)[-\sin(\sec^2 2t)]\times\frac{d(\sec^2 2t)}{dt}$ $\Rightarrow \frac{dy}{dt} = -3\cos^2(\sec^2 2t).\sin(\sec^2 2t)\times 2\sec 2t.\sec 2t\tan 2t.2$ $\therefore \frac{dy}{dt} = -12\cos^2(\sec^2 2t)\times\sin(\sec^2 2t)\times\sec^2 2t\times\tan 2t.$	½ 1 ½
	OR	
22(b).	If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.	
Sol.	$As, x^y = e^{x-y} \Rightarrow \log(x^y) = \log(e^{x-y})$ $\Rightarrow y\log x = (x-y) \Rightarrow y = \frac{x}{1+\log x}$ *Now*, Differentiating both the sides wrt x $\frac{dy}{dx} = \frac{(\log x+1).1-x(\frac{1}{x})}{(\log x+1)^2} = \frac{\log x}{(1+\log x)^2}$	1 1
23.	Find the interval in which the function $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing.	

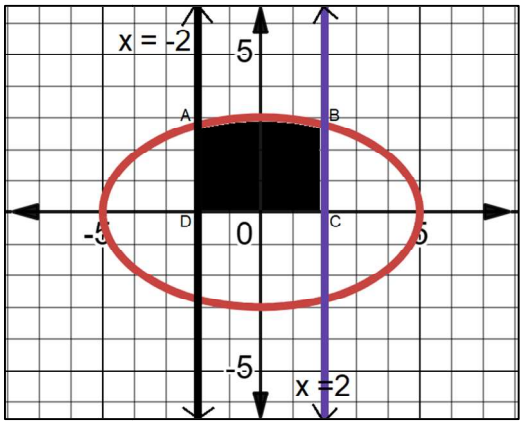
Sol.	$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$ $\Rightarrow 4x^2(x-3) < 0$ for $x < 3, x \neq 0$ $\Rightarrow f'(x) < 0$ for $x < 3, x \neq 0$ Thus, $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing on $(-\infty, 0) \cup (0,3)$. OR Thus, $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing on $(-\infty, 0] \cup [0,3]$	**1** **½** **½**
24.	The volume of a cube is increasing at the rate of 6 cm^3/s. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?	
Sol.	Given, $\frac{dV}{dt} = 6 \text{ cm}^3/\text{sec}$. Since, $V = x^3$ $\frac{dV}{dt} = 3x^2 \frac{dx}{dt} \Rightarrow 6 = 3x^2 \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{2}{x^2}$ cm / sec Now, Surface Area = S = $6x^2 \Rightarrow \frac{dS}{dt} = 12x\frac{dx}{dt} = 3 \text{ cm}^2/\text{sec}$	**1** **1**
25.	Find: $\int \frac{dx}{x(x^2-1)}$	
Sol.	$I = \int \frac{dx}{x(x^2-1)} = \int \frac{dx}{x^3(1-\frac{1}{x^2})} = \frac{1}{2}\int \frac{\left(\frac{2}{x^3}\right)dx}{(1-\frac{1}{x^2})}$ Put $(1-\frac{1}{x^2}) = t \Rightarrow \left(\frac{2}{x^3}\right)dx = dt$ $I = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log\|(1-\frac{1}{x^2})\| + c$ OR $\frac{1}{2}\log\|(\frac{x^2-1}{x^2})\| + c$	**1** **1**
	SECTION C **In this section there are 6 short answer type questions of 3 marks each.**	
26.	Given that $y = (\sin x)^x . x^{\sin x} + a^x$, find $\frac{dy}{dx}$.	
Sol.	As, $y = (\sin x)^x . x^{\sin x} + a^x = u + a^x \Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{d(a^x)}{dx}$ where $u = (\sin x)^x . x^{\sin x} \Rightarrow \log u = x\log(\sin x) + \sin x . \log x$ on differentiating both sides with respect to x, we get $\Rightarrow \frac{du}{dx} = (\sin x)^x . x^{\sin x}[\log(\sin x) + x\cot x + \frac{\sin x}{x} + \log x . \cos x]$ Thus, $\frac{dy}{dx} = (\sin x)^x . x^{\sin x}[\log(\sin x) + x\cot x + \frac{\sin x}{x} + \log x . \cos x] + a^x \log a$	**½** **1 ½** **1**

27(a).	Evaluate : $\int_0^{\frac{\pi}{4}} \frac{x}{1+\cos 2x+\sin 2x}dx$	
Sol.	$I=\int_0^{\frac{\pi}{4}} \frac{x}{1+\cos 2x+\sin 2x}dx$ ....(1) *On* applying $\int_0^a f(x)dx=\int_0^a f(a-x)dx,$ we get $I=\int_0^{\frac{\pi}{4}} \frac{\frac{\pi}{4}-x}{1+\cos 2x+\sin 2x}dx$...(2) *On* adding Eq. (1) and (2), we get $2I=\frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{1}{1+\cos 2x+\sin 2x}dx$ $I=\frac{\pi}{16}\int_0^{\frac{\pi}{4}} \frac{1}{\cos^2 x+\sin x\cos x}dx=\frac{\pi}{16}\int_0^{\frac{\pi}{4}} \frac{\sec^2 xdx}{1+\tan x}$ $I=\frac{\pi}{16}(\log\lvert 1+\tan x\rvert)_0^{\frac{\pi}{4}}$ $I=\frac{\pi}{16}\log 2$	½ ½ **1** ½ ½
	OR	
27(b).	Find : $\int e^x\left[\frac{1}{(1+x^2)^{\frac{3}{2}}}+\frac{x}{\sqrt{1+x^2}}\right]dx$	
Sol.	$I=\int e^x(\frac{x}{\sqrt{1+x^2}}+\frac{1}{(1+x^2)^{\frac{3}{2}}})dx$ Let $f(x)=\frac{x}{\sqrt{1+x^2}}, f'(x)=\frac{\sqrt{1+x^2}-x\frac{x}{\sqrt{1+x^2}}}{1+x^2}=\frac{1}{(1+x^2)^{\frac{3}{2}}}$ On applying $\int e^x(f(x)+f'(x))dx=e^x f(x)+c,$ $I=e^x\frac{x}{\sqrt{1+x^2}}+c$	½ +1½ **1**
28.	Find: $\int \frac{3x+5}{\sqrt{x^2+2x+4}}dx$	
Sol.	$I=\int \frac{3x+5}{\sqrt{x^2+2x+4}}dx=\frac{3}{2}\int \frac{2x+2}{\sqrt{x^2+2x+4}}dx+2\int \frac{1}{\sqrt{x^2+2x+4}}dx$ $I=\frac{3}{2}(2\sqrt{x^2+2x+4})+2\int \frac{1}{\sqrt{(x+1)^2+\sqrt{3}^2}}dx$ $I=3\sqrt{x^2+2x+4}+2\log\lvert(x+1)+\sqrt{x^2+2x+4}\rvert+c$	**1** **1+** ½ ½

29(a).	Find the particular solution of the differential equation $\frac{dy}{dx} = y \cot 2x$, given that $y\left(\frac{\pi}{4}\right) = 2$.	
Sol.	$\frac{dy}{dx} = y\cot 2x \Rightarrow \int \frac{dy}{y} = \int \cot 2x \, dx$ $\Rightarrow \log \lvert y \rvert = \frac{1}{2}\log \lvert \sin 2x \rvert + \log c$ $y = c.\sqrt{\sin 2x}$ when $y(\frac{\pi}{4}) = 2$, gives $c = 2$ $\therefore y = 2\sqrt{\sin 2x}$ is the required Particular solution of given D.E.	**1** **1** ½ ½
	OR	
29(b).	Find the particular solution of the differential equation $(xe^{\frac{y}{x}} + y)\,dx = x\,dy$, given that y = 1 when $x = 1$.	
Sol.	$\frac{dy}{dx} = e^{\frac{y}{x}} + \frac{y}{x} = f(\frac{y}{x})$ so, its a homogeneous differential equation Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ Now, $v + x\frac{dv}{dx} = e^v + v$ $\Rightarrow \int e^{-v} dv = \int \frac{1}{x} dx$ $\Rightarrow -e^{-v} = \log \lvert x \rvert + c \Rightarrow -e^{\frac{-y}{x}} = \log \lvert x \rvert + c$.....(1) Now, $x = 1$, $y = 1$,gives $c = -e^{-1}$ Thus, $\boxed{\log \lvert x \rvert + e^{\frac{-y}{x}} = e^{-1}}$	**1** ½ **1** ½
30.	Solve the following linear programming problem graphically: Maximise $Z = 2x + 3y$ subject to the constraints: $x + y \le 6, x \ge 2,\ y \le 3, \& x \ge 0, y \ge 0$	

Sol.	On plotting the graph of $x+y\le 6, x\ge 2,\ y\le 3, \& x\ge 0, y\ge 0$ we get the following graph and common shaded region is the region ABCD.  Now, Corner points of the common shaded region are $A(2,3), B(3,3), C(6,0) \& D(2,0)$.Thus, Corner points | Value of $Z=2x+3y$ $A(2,3)$ | 13 $B(3,3)$ | 15 $C(6,0)$ | 12 $D(2,0)$ | 4 So, Maximum Value of Z is 15 at $x=3,\ y=3$.	**For correct Graph 1 ½** **For correct Table 1** **½**
31(a).	A card from a well shuffled deck of 52 playing cards is lost. From the remaining cards of the pack, a card is drawn at random and is found to be a King. Find the probability of the lost card being a King.	
Sol.	Let E_1 be the event of lost card is King, E_2 be the event of lost card not a King and A be the event of drawing a King from remaining 51 cards. so, $P(E_1)=\frac{1}{13}, P(E_2)=\frac{12}{13}, P(A\|E_1)=\frac{3}{51}, P(A\|E_2)=\frac{4}{51}$ Now, Required probability is $P(E_1\|A)$, $P(E_1\|A)=\frac{P(A\|E_1)\times P(E_1)}{P(A\|E_1)\times P(E_1)+P(A\|E_2)\times P(E_2)}=\frac{\frac{1}{13}\times\frac{3}{51}}{\frac{1}{13}\times\frac{3}{51}+\frac{12}{13}\times\frac{4}{51}}=\frac{1}{17}$	**½** **1 ½** **1**
	OR	

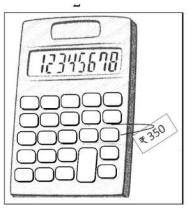
31(b).	A biased die is twice as likely to show an even number as an odd number. If such a die is thrown twice, find the probability distribution of the number of sixes. Also, find the mean of the distribution.	
Sol.	Let P(1)=P(3)=P(5) = p, so P(2)=P(4)=P(6) =2 p As, P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1 $\Rightarrow$ 9p = 1 $\Rightarrow$ p = $\frac{1}{9}$ P(Getting 6)=$\frac{2}{9}$,P(Not getting six)=$\frac{7}{9}$	½
	Let X represents the Number of sixes Possible values of X are 0, 1 or 2	½
	Now, P(X=0)=$\frac{7}{9}\times\frac{7}{9}=\frac{49}{81}$, P(X=1)=$2\times\frac{7}{9}\times\frac{2}{9}=\frac{28}{81}$, P(X=2)=$\frac{2}{9}\times\frac{2}{9}=\frac{4}{81}$ Required probability distribution of number of sixes is $\begin{array}{\|c\|ccc\|}\hline X & 0 & 1 & 2 \\ \hline P(X) & \frac{49}{81} & \frac{28}{81} & \frac{4}{81} \\ \hline \end{array}$	1½
	Mean of X = $\sum_{i=1}^{3} X_i P(X_i) = 0+\frac{28}{81}+\frac{8}{81}=\frac{4}{9}$	½
	SECTION D **In this section there are 4 long answer type questions of 5 marks each.**	
32(a).	Sketch the graph of y = $x\|x\|$ and hence find the area bounded by this curve, X-axis and the ordinates x = –2 and x = 2, using integration.	
Sol.	 As, $y = x\|x\| = \begin{cases} -x^2, x<0 \\ x^2, x\geq 0 \end{cases}$	**For correct figure 1 ½**

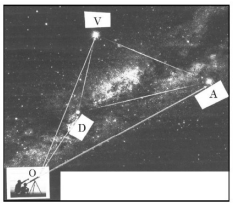
	Area of the shaded region $= \int_{-2}^{2} y\,dx = 2\int_{0}^{2} y\,dx = 2\int_{0}^{2} x^2\,dx$	1 ½
	$= 2\left(\frac{x^3}{3}\right)_0^2$	1
	$= 2\left(\frac{8}{3}\right) = \frac{16}{3}$	1
	OR	
32(b).	Using integration, find the area bounded by the ellipse $9x^2 + 25y^2 = 225$, the lines $x = -2$, $x = 2$, and the X-axis.	
Sol.		**For correct figure 1 ½**
	As, $9x^2 + 25y^2 = 225 \Rightarrow \boxed{y = \pm\frac{3}{5}\sqrt{5^2 - x^2}}$	
	Required Area $= \int_{-2}^{2} \frac{3}{5}\sqrt{5^2 - x^2}\,dx = \frac{6}{5}\int_{0}^{2} \sqrt{5^2 - x^2}\,dx$	1 ½
	$= \frac{6}{5}\left(\frac{x\sqrt{5^2 - x^2}}{2} + \frac{25}{2}\sin^{-1}\left(\frac{x}{5}\right)\right)_0^2$	1
	$= \frac{6}{5}\left(\frac{2\sqrt{21}}{2} + \frac{25}{2}\sin^{-1}\left(\frac{2}{5}\right)\right)$	
	$= \left(\frac{6\sqrt{21}}{5} + 15\sin^{-1}\left(\frac{2}{5}\right)\right)$	1

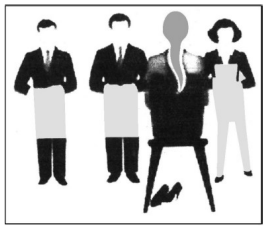
33(a).	Let A = R – {5} and B = R – {1}. Consider the function f : A → B, defined by $f(x) = \frac{x-3}{x-5}$. Show that f is one-one and onto.	
Sol.	Let $f(x_1) = f(x_2)$, for some $x_1, x_2 \in A$ $\Rightarrow \frac{x_1-3}{x_1-5} = \frac{x_2-3}{x_2-5}$ $\Rightarrow (x_1-3)(x_2-5) = (x_2-3)(x_1-5)$ $\Rightarrow x_1 = x_2$, So f *is one – one Function.* Let $y = f(x) = \frac{x-3}{x-5} \Rightarrow y(x-5) = x-3$ $\Rightarrow yx - 5y = x - 3$ $\Rightarrow x = \frac{5y-3}{y-1}$, We observe that x is defined for all values of y except $y = 1$, So, Range = $R - \{1\}$ and Co-domain is Given $R - \{1\}$ [As, $f : A \to B$] Since, Range = Co-domain, f is onto Function. Thus, f is one-one & onto function.	**2 ½** **2 ½**
	OR	
33(b).	Check whether the relation S in the set of real numbers R defined by S = {(a, b) : where a – b + $\sqrt{2}$ is an irrational number} is reflexive, symmetric or transitive.	
Sol.	Reflexive: For $a \in S$ $\Rightarrow a - a + \sqrt{2}$ is irrational number $\Rightarrow \sqrt{2}$ is irrational number $\Rightarrow (a, a) \in S$ Thus, S is Reflexive Relation. Symmetric : Let $(a,b) \in S \Rightarrow a - b + \sqrt{2}$ is irrational number but $b - a + \sqrt{2}$ may not be irrational number For example, $(\sqrt{2}, 1) \in S \Rightarrow \sqrt{2} - 1 + \sqrt{2} = 2\sqrt{2} - 1$ is irrational number $(1, \sqrt{2}) \notin S$ as $1 - \sqrt{2} + \sqrt{2} = 1$ is not irrational number $\therefore (b, a) \notin S$, So S is NOT Symmetric Relation.	**1 ½** **1 ½**

	Transitive : *Let* $(a,b)\in S \Rightarrow a-b+\sqrt{2}$ is irrational number $\&(b,c)\in S \Rightarrow b-c+\sqrt{2}$ is irrational number but $a-c+\sqrt{2}$ may not be irrational number *For* example, $(1,\sqrt{3})\in S \Rightarrow 1-\sqrt{3}+\sqrt{2}$ is irrational number $(\sqrt{3},\sqrt{2})\in S \Rightarrow \sqrt{3}-\sqrt{2}+\sqrt{2}=\sqrt{3}$ is irrational number But $(1,\sqrt{2})\notin S$ as $1-\sqrt{2}+\sqrt{2}=1$ is not irrational number $\therefore (a,c)\notin S$, *So S* is NOT Transitive Relation. Thus, S is Reflexive But Neither Symmetric nor Transitive Relation.	**2**
34.	If $A=\begin{bmatrix}2 & 1 & -3\\ 3 & 2 & 1\\ 1 & 2 & -1\end{bmatrix}$, find A^{-1} and hence solve the following system of equations : $2x + y - 3z = 13$ $3x + 2y + z = 4$ $x + 2y - z = 8$	
Sol.	*For* Matrix $A=\begin{pmatrix}2 & 1 & -3\\ 3 & 2 & 1\\ 1 & 2 & -1\end{pmatrix}$, $\lvert A\rvert=-16\neq 0$ so, A^{-1} exists.	**1**
	$adjA=\begin{pmatrix}-4 & -5 & 7\\ 4 & 1 & -11\\ 4 & -3 & 1\end{pmatrix}$,	**1 ½**
	Thus, $A^{-1}=\frac{-1}{16}\begin{pmatrix}-4 & -5 & 7\\ 4 & 1 & -11\\ 4 & -3 & 1\end{pmatrix}$	**½**
	so, Given equation can be written into a matrix equation as $\begin{pmatrix}2 & 1 & -3\\ 3 & 2 & 1\\ 1 & 2 & -1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}13\\ 4\\ 8\end{pmatrix}\Rightarrow X=A^{-1}.B$	**½**
	A X = B $\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\frac{-1}{16}\begin{pmatrix}-4 & -5 & 7\\ 4 & 1 & -11\\ 4 & -3 & 1\end{pmatrix}\begin{pmatrix}13\\ 4\\ 8\end{pmatrix}=\frac{-1}{16}\begin{pmatrix}-16\\ -32\\ 48\end{pmatrix}=\begin{pmatrix}1\\ 2\\ -3\end{pmatrix}\Rightarrow x=1, y=2, z=-3$	**1 ½**

35(a).	Find the distance between the line $\frac{x}{2}=\frac{2y-6}{4}=\frac{1-z}{-1}$ and another line parallel to it passing through the point (4, 0, –5).	
Sol.	Equation of the given line in standard form is $L_1:\frac{x}{2}=\frac{y-3}{2}=\frac{z-1}{1}$ Equation of the line parallel to L_1 & passing through $(4,\ 0,-5)$ is $L_2:\frac{x-4}{2}=\frac{y}{2}=\frac{z+5}{1}$ Vector Equation of Lines are $L_1:\vec{r}=(0\hat{i}+3\hat{j}+\hat{k})+\lambda(2\hat{i}+2\hat{j}+\hat{k})$ $L_2:\vec{r}=(4\hat{i}+0\hat{j}-5\hat{k})+\mu(2\hat{i}+2\hat{j}+\hat{k})$ Now, $\overrightarrow{a_2}-\overrightarrow{a_1}=(4\hat{i}+0\hat{j}-5\hat{k})-(0\hat{i}+3\hat{j}+\hat{k})=(4\hat{i}-3\hat{j}-6\hat{k})$ $\vec{b}=2\hat{i}+2\hat{j}+\hat{k}$ $(\overrightarrow{a_2}-\overrightarrow{a_1})\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\4&-3&-6\\2&2&1\end{vmatrix}=9\hat{i}-16\hat{j}+14\hat{k}$ $\lvert\vec{b}\rvert=\sqrt{4+4+1}=3$ Thus, distance between the lines is S.D. $=\frac{\lvert(\overrightarrow{a_2}-\overrightarrow{a_1})\times\vec{b}\rvert}{\lvert\vec{b}\rvert}=\frac{\sqrt{81+256+196}}{3}=\frac{\sqrt{533}}{3}$ units	½ **1** ½ **1** ½ **1 ½**
	OR	
35(b).	If the lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-7}$ are perpendicular to each other, find the value of k and hence write the vector equation of a line perpendicular to these two lines and passing through the point (3, –4, 7).	
	$L_1:\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}\Rightarrow$ direction ratio's of $L_1=<-3,2k,2>$ $L_2:\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-7}\Rightarrow$ direction ratio's of $L_2=<3k,1,-7>$ Since $L_1\perp L_2$, $-9k+2k-14=0\Rightarrow k=-2$	½ ½ **1**

	Thus, d.r.'s of $L_1 =< -3,-4,2 >$, d.r.'s of $L_2 =< -6,1,-7 >$ Now the vector perpendicular to both L_1 & L_2 is given by $\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & -4 & 2 \\ -6 & 1 & -7 \end{vmatrix} = 26\hat{i} - 33\hat{j} - 27\hat{k}$ Thus, Equation of the required line is $\vec{r} = (3\hat{i} - 4\hat{j} + 7\hat{k}) + \lambda(26\hat{i} - 33\hat{j} - 27\hat{k})$	**2** **1**
	SECTION E **In this section there are 3 case-study based questions of 4 marks each.**	
36.	A store has been selling calculators at Rs. 350 each. A market survey indicates that a reduction in price (p) of calculator increases the number of units (x) sold. The relation between the price and quantity sold is given by demand function $p = 450 - \frac{x}{2}$.  Based on the above information, answer the following questions: (i) Determine the number of units (x) that should be sold to maximise the revenue $R(x) = xp(x)$. Also verify the result. (ii) What rebate in price of calculator should the store give to maximise the revenue?	
Sol.	(i) Revenue by selling x items = $R(x) = x.p(x) = 450x - \frac{x^2}{2}$ $\frac{dR}{dx} = 450 - x$ For Maxima or Minima, $\frac{dR}{dx} = 0 \Rightarrow x = 450$ $\frac{d^2R}{dx^2} = -1 < 0$ (Revenue is maximum when x = 450 units are sold) (ii) At $x = 450$, $p = 450 - \frac{450}{2} = 225$ So, Rebate = $350 - 225$ = Rs.125 per calculator	**½** **1** **½** **1** **1**

37.	An instructor at the astronomical centre shows three among the brightest stars in a particular constellation. Assume that the telescope is located at O(0,0,0) and the three stars have their locations at the points D, A and V having position vectors $2\hat{i}+3\hat{j}+4\hat{k}, 7\hat{i}+5\hat{j}+8\hat{k}$ and $-3\hat{i}+7\hat{j}+11\hat{k}$ respectively.  Based on the above information, answer the following questions: (i) How far is the star V from star A? (ii) Find a unit vector in the direction of $\overrightarrow{DA}$. (iii) Find the measure of $\angle$VDA. OR What is the projection of vector $\overrightarrow{DV}$ on vector $\overrightarrow{DA}$?	 1 1 2 2
Sol.	(i) $\overrightarrow{AV}$ = Position Vector of V − Position Vector of A $= -3\hat{i}+7\hat{j}+11\hat{k}-7\hat{i}-5\hat{j}-8\hat{k} = -10\hat{i}+2\hat{j}+3\hat{k}$ Thus, $\lvert\overrightarrow{AV}\rvert = \sqrt{100+4+9} = \sqrt{113}$ units (ii) $\overrightarrow{DA}$ = Position Vector of A − Position Vector of D $= 7\hat{i}+5\hat{j}+8\hat{k}-2\hat{i}-3\hat{j}-4\hat{k} = 5\hat{i}+2\hat{j}+4\hat{k}$ Unit vector in the direction of $\overrightarrow{DA} = \frac{5\hat{i}+2\hat{j}+4\hat{k}}{3\sqrt{5}}$ (iii) $\overrightarrow{DV} = -5\hat{i}+4\hat{j}+7\hat{k}$ $\angle VDA = \cos^{-1}\left(\frac{\overrightarrow{DV}.\overrightarrow{DA}}{\lvert\overrightarrow{DV}\rvert\lvert\overrightarrow{DA}\rvert}\right) = \cos^{-1}\left(\frac{11\sqrt{2}}{90}\right)$ *OR* (iii) $\overrightarrow{DV} = -5\hat{i}+4\hat{j}+7\hat{k}$ Projection of $\overrightarrow{DV}$ on $\overrightarrow{DA} = \left(\frac{\overrightarrow{DV}.\overrightarrow{DA}}{\lvert\overrightarrow{DA}\rvert}\right) = \frac{11\sqrt{5}}{15}$	 ½ ½ ½ ½ ½ **1 ½** ½ **1 ½**
38.	Rohit, Jaspreet and Alia appeared for an interview for three vacancies in the same post. The probability of Rohit's selection is $\frac{1}{5}$, Jaspreet's selection is $\frac{1}{3}$ and Alia's selection is $\frac{1}{4}$. The event of selection is independent of each other.	

	 Based on the above information, answer the following questions: (i) What is the probability that at least one of them is selected ? (ii) Find P(G|$\overline{H}$) where G is the event of Jaspreet's selection and $\overline{H}$ denotes the event that Rohit is not selected. (iii) Find the probability that exactly one of them is selected. OR (iii) Find the probability that exactly two of them are selected.	**1** **1** **2** **2**
Sol.	Given P(Rohit) = $\frac{1}{5}$, P(Jaspreet) = $\frac{1}{3}$, P(Alia) = $\frac{1}{4}$ (*i*) P(atleast one of them is selected) = 1 – P(no one is selected) $= 1-\left(\frac{4}{5}\times\frac{2}{3}\times\frac{3}{4}\right)=\frac{3}{5}$ (*ii*) P(G|$\overline{H}$) = $\frac{P(G\cap\overline{H})}{P(\overline{H})}=\frac{1}{3}$ (*iii*) P(exactly one of them selected) $= P(R)\times P(\overline{J})\times P(\overline{A})+P(\overline{R})\times P(J)\times P(\overline{A})+P(\overline{R})\times P(\overline{J})\times P(A)$ $=\frac{6+12+8}{60}=\frac{13}{30}$ OR (*iii*) P(exactly two of them selected) $= P(R)\times P(J)\times P(\overline{A})+P(R)\times P(\overline{J})\times P(A)+P(\overline{R})\times P(J)\times P(A)$ $=\frac{3+2+4}{60}=\frac{3}{20}$	**1** **1** **1** **1** **1** **1**

MARKING SCHEME – 65/5/1

Q.No.	EXPECTED ANSWER / VALUE POINTS	Marks
	SECTION-A **(Question nos. 1 to 18 are Multiple choice Questions carrying 1 mark each)**	
1.	A function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = x^2 - 4x + 5$ is : (A) injective but not surjective. (B) surjective but not injective. (C) both injective and surjective. (D) neither injective nor surjective.	
Ans	**(D) neither injective nor surjective**	**1**
2.	If $A = \begin{bmatrix} a & c & -1 \\ b & 0 & 5 \\ 1 & -5 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the value of $2a - (b + c)$ is : (A) 0 (B) 1 (C) –10 (D) 10	
Ans	**(A) 0**	**1**
3.	If A is a square matrix of order 3 such that the value of $\lvert \text{adj} \cdot A \rvert = 8$, then the value of $\lvert A^T \rvert$ is : (A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 8 (D) $2\sqrt{2}$	
Ans	**(D) $2\sqrt{2}$**	**1**
4.	If inverse of matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 1 & \lambda & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then value of λ is : (A) –4 (B) 1 (C) 3 (D) 4	
Ans	**(D) 4**	**1**

5.	If $\begin{bmatrix} x & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ x \end{bmatrix} = \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ x \end{bmatrix}$, then value of x is : (A) -1 (B) 0 (C) 1 (D) 2	
Ans	**(A) –1**	**1**
6.	Find the matrix A^2, where $A = [a_{ij}]$ is a 2×2 matrix whose elements are given by a_{ij} = maximum (i, j) – minimum (i, j) : (A) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	
Ans	**(C)** $\begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}$	**1**
7.	If $xe^y = 1$, then the value of $\frac{dy}{dx}$ at $x = 1$ is : (A) -1 (B) 1 (C) $-e$ (D) $-\frac{1}{e}$	
Ans	**(A) –1**	**1**
8.	Derivative of $e^{\sin^2 x}$ with respect to $\cos x$ is : (A) $\sin x\, e^{\sin^2 x}$ (B) $\cos x\, e^{\sin^2 x}$ (C) $-2 \cos x\, e^{\sin^2 x}$ (D) $-2 \sin^2 x \cos x\, e^{\sin^2 x}$	
Ans	**(C)** $\mathbf{-2 \cos x\, e^{\sin^2 x}}$	**1**
9.	The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minima at x equal to : (A) 2 (B) 1 (C) 0 (D) –2	
Ans	**(A) 2**	**1**

10.	Given a curve $y = 7x - x^3$ and x increases at the rate of 2 units per second. The rate at which the slope of the curve is changing, when $x = 5$ is : (A) –60 units/sec (B) 60 units/sec (C) –70 units/sec (D) –140 units/sec	
Ans	**(A) –60 units/sec**	**1**
11.	$\int \frac{1}{x(\log x)^2}\,dx$ is equal to : (A) $2 \log (\log x) + c$ (B) $-\frac{1}{\log x} + c$ (C) $\frac{(\log x)^3}{3} + c$ (D) $\frac{3}{(\log x)^3} + c$	
Ans	**(B)** $\mathbf{-\frac{1}{\log x} + c}$	**1**
12.	The value of $\int_{-1}^{1} x\,\lvert x\rvert\,dx$ is : (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $-\frac{1}{6}$ (D) 0	
Ans	**(D) 0**	**1**
13.	Area of the region bounded by curve $y^2 = 4x$ and the X-axis between $x = 0$ and $x = 1$ is : (A) $\frac{2}{3}$ (B) $\frac{8}{3}$ (C) 3 (D) $\frac{4}{3}$	
Ans	**(D)** $\mathbf{\frac{4}{3}}$	**1**
14.	The order of the differential equation $\frac{d^4y}{dx^4} - \sin\left(\frac{d^2y}{dx^2}\right) = 5$ is : (A) 4 (B) 3 (C) 2 (D) not defined	
Ans	**(A) 4**	**1**

15.	The position vectors of points P and Q are $\vec{p}$ and $\vec{q}$ respectively. The point R divides line segment PQ in the ratio 3 : 1 and S is the mid-point of line segment PR. The position vector of S is : (A) $\frac{\vec{p}+3\vec{q}}{4}$ (B) $\frac{\vec{p}+3\vec{q}}{8}$ (C) $\frac{5\vec{p}+3\vec{q}}{4}$ (D) $\frac{5\vec{p}+3\vec{q}}{8}$	
Ans	**(D)** $\mathbf{\frac{5\vec{p}+3\vec{q}}{8}}$	**1**
16.	The angle which the line $\frac{x}{1}=\frac{y}{-1}=\frac{z}{0}$ makes with the positive direction of Y-axis is : (A) $\frac{5\pi}{6}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{7\pi}{4}$	
Ans	**(B)** $\mathbf{\frac{3\pi}{4}}$	**1**
17.	The Cartesian equation of the line passing through the point (1, –3, 2) and parallel to the line : $\vec{r}=(2+\lambda)\hat{i}+\lambda\hat{j}+(2\lambda-1)\hat{k}$ is (A) $\frac{x-1}{2}=\frac{y+3}{0}=\frac{z-2}{-1}$ (B) $\frac{x+1}{1}=\frac{y-3}{1}=\frac{z+2}{2}$ (C) $\frac{x+1}{2}=\frac{y-3}{0}=\frac{z+2}{-1}$ (D) $\frac{x-1}{1}=\frac{y+3}{1}=\frac{z-2}{2}$	
Ans	**(D)** $\mathbf{\frac{x-1}{1}=\frac{y+3}{1}=\frac{z-2}{2}}$	**1**
18.	If A and B are events such that P(A/B) = P(B/A) ≠ 0, then : (A) $A \subset B$, but $A \neq B$ (B) $A = B$ (C) $A \cap B = \phi$ (D) $P(A) = P(B)$	
Ans	**(D)** $\mathbf{P(A)=P(B)}$	**1**

	(Question Nos. 19 & 20 are Assertion-Reason based questions of 1 mark each)	
	Assertion – Reason Based Questions **Direction :** In questions numbers **19** and **20**, two statements are given one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the following options : (A) Both Assertion (A) and Reason (R) are true and the Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	
19.	**Assertion (A) :** Domain of $y = \cos^{-1}(x)$ is $[-1, 1]$. **Reason (R) :** The range of the principal value branch of $y = \cos^{-1}(x)$ is $[0,\pi]-\left\{\frac{\pi}{2}\right\}$.	
Ans	**(C) Assertion (A) is true, but Reason (R) is false**	**1**
20.	**Assertion (A) :** The vectors $\vec{a} = 6\hat{i} + 2\hat{j} - 8\hat{k}$ $\vec{b} = 10\hat{i} - 2\hat{j} - 6\hat{k}$ $\vec{c} = 4\hat{i} - 4\hat{j} + 2\hat{k}$ represent the sides of a right angled triangle. **Reason (R) :** Three non-zero vectors of which none of two are collinear forms a triangle if their resultant is zero vector or sum of any two vectors is equal to the third.	
Ans	**(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A)**	**1**
	SECTION-B **(Question nos. 21 to 25 are very short Answer type questions carrying 2 marks each)**	
21.	Find value of k if $\sin^{-1}\left[k\tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right] = \frac{\pi}{3}$.	

Ans	$k\tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)=\sin\frac{\pi}{3}$	$\frac{1}{2}$
	$\Rightarrow\ k\tan\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{2}$	$\frac{1}{2}+\frac{1}{2}$
	$\Rightarrow\ k\sqrt{3}=\frac{\sqrt{3}}{2}\ \therefore k=\frac{1}{2}$	$\frac{1}{2}$
22.	(a) Verify whether the function f defined by $f(x)=\begin{cases}x\sin\left(\frac{1}{x}\right), & x\neq 0\\ 0, & x=0\end{cases}$ is continuous at $x=0$ or not. **OR** (b) Check for differentiability of the function f defined by $f(x)=\lvert x-5\rvert$, at the point $x=5$.	
Ans	**(a)** $\lim_{x\to 0} f(x)=\lim_{x\to 0} x.\sin\frac{1}{x}=0\times$ **Finite value in** $[-1,1]=0=f(0)$	$1\frac{1}{2}$
	$\therefore$ **f is a continuous function.**	$\frac{1}{2}$
	Or	
	(b) $\text{LHD}=\lim_{x\to 5^-}\frac{\lvert x-5\rvert-0}{x-5}=\lim_{x\to 5^-}\frac{-(x-5)}{x-5}=-1$	1
	$\text{RHD}=\lim_{x\to 5^+}\frac{\lvert x-5\rvert-0}{x-5}=\lim_{x\to 5^+}\frac{(x-5)}{x-5}=1$	$\frac{1}{2}$
	$\text{LHD}\neq\text{RHD},\ \therefore f$ **is not differentiable at** $x=5$	$\frac{1}{2}$
23.	The area of the circle is increasing at a uniform rate of 2 cm^2/sec. How fast is the circumference of the circle increasing when the radius r = 5 cm ?	
Ans	**Let** $C=2\pi r$**, be the circumference of the circle,**	
	$\frac{d(\pi r^2)}{dt}=2\Rightarrow\frac{dr}{dt}=\frac{1}{\pi r}$ cm / sec	1
	$\frac{dC}{dt}=2\pi\frac{dr}{dt}=\frac{2}{r}=\frac{2}{5}$ cm/sec at $r=5$ cm	1

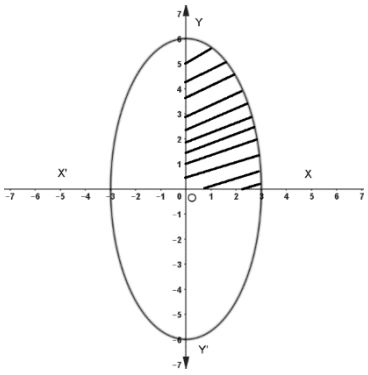
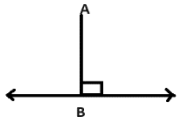
24.	(a) Find : $\int \cos^3 x\, e^{\log \sin x}\, dx$ **OR** (b) Find : $\int \frac{1}{5 + 4x - x^2}\, dx$	
Ans	**(a)** $\int \cos^3 x\, e^{\log \sin x} dx = \int \cos^3 x \cdot \sin x\, dx$, **Assuming** $\cos x = t$ **and** $\sin x\, dx = -dt$ $= -\int t^3 dt$ $= -\frac{t^4}{4} + C = -\frac{\cos^4 x}{4} + C$ **Or** **(b)** $\int \frac{1}{5+4x-x^2} dx = \int \frac{1}{3^2 - (x-2)^2} dx$ $= \frac{1}{6} \log \left\lvert \frac{1+x}{5-x} \right\rvert + C$	**1** **½** **½** **1** **1**
25.	Find the vector equation of the line passing through the point (2, 3, –5) and making equal angles with the co-ordinate axes.	
Ans	**(a)** $\cos\alpha = \cos\beta = \cos\gamma = l \Rightarrow l^2 + l^2 + l^2 = 1 \Rightarrow 3l^2 = 1$, $\therefore l = 1/\sqrt{3}$ **Direction cosines of the line are** $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \Rightarrow$ **the direction ratios are 1,1,1** $\therefore$ **Vector equation of the line is:** $\vec{r} = 2\hat{i} + 3\hat{j} - 5\hat{k} + \lambda(\hat{i} + \hat{j} + \hat{k})$	**1** **½** **½**
	SECTION-C **(Question nos. 26 to 31 are short Answer type questions carrying 3 marks each)**	
26.	(a) Find $\frac{dy}{dx}$, if $(\cos x)^y = (\cos y)^x$. **OR** (b) If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.	
Ans	**(a) Taking 'log' on both sides of** $(\cos x)^y = (\cos y)^x$**, we get** $y \log \cos x = x \log \cos y$	**1**

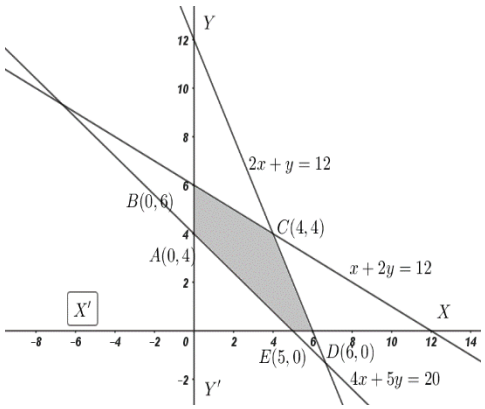
	$\Rightarrow \frac{dy}{dx}\log\cos x + y(-\tan x) = \log\cos y + x(-\tan y)\frac{dy}{dx}$	$1\frac{1}{2}$
	$\Rightarrow \frac{dy}{dx} = \frac{\log\cos y + y\tan x}{\log\cos x + x\tan y}$	$\frac{1}{2}$
	Or	
	(b) Let $x = \sin A, y = \sin B \therefore A = \sin^{-1}x,\ B = \sin^{-1}y$	$\frac{1}{2}$
	$\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$	
	$\Rightarrow \cos A + \cos B = a(\sin A - \sin B)$	
	$\Rightarrow 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) = 2a\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$	1
	$\Rightarrow \cot\left(\frac{A-B}{2}\right) = a \Rightarrow A - B = 2\cot^{-1}a$, $\therefore \sin^{-1}x - \sin^{-1}y = 2\cot^{-1}a$, differentiating with respect to 'x'.	$\frac{1}{2}$
	$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$	1
27.	If $x = a\sin^3\theta$, $y = b\cos^3\theta$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.	
Ans	$\frac{dx}{d\theta} = 3a\sin^2\theta\cos\theta,\ \frac{dy}{d\theta} = -3b\cos^2\theta\sin\theta$	1
	$\Rightarrow \frac{dy}{dx} = \frac{-3b\cos^2\theta\sin\theta}{3a\sin^2\theta\cos\theta} = -\frac{b}{a}\cot\theta$	$\frac{1}{2}$
	$\Rightarrow \frac{d^2y}{dx^2} = \frac{b}{a}\text{cosec}^2\theta\frac{d\theta}{dx} = \frac{b}{a}\text{cosec}^2\theta\cdot\frac{1}{3a\sin^2\theta\cos\theta} = \frac{b}{3a^2}\sec\theta\,\text{cosec}^4\theta$	1
	$\Rightarrow \left.\frac{d^2y}{dx^2}\right]_{\theta=\frac{\pi}{4}} = \frac{4\sqrt{2}b}{3a^2}$	$\frac{1}{2}$
28.	(a) Evaluate : $\int_0^{\pi}\frac{e^{\cos x}}{e^{\cos x}+e^{-\cos x}}dx$	
	OR	
	(b) Find : $\int\frac{2x+1}{(x+1)^2(x-1)}dx$	
Ans	**(a)** Let $I = \int_0^{\pi}\frac{e^{\cos x}}{e^{\cos x}+e^{-\cos x}}dx$ --------- (i)	
	$\Rightarrow I = \int_0^{\pi}\frac{e^{\cos(\pi-x)}}{e^{\cos(\pi-x)}+e^{-\cos(\pi-x)}}dx = \int_0^{\pi}\frac{e^{-\cos x}}{e^{-\cos x}+e^{\cos x}}dx$ --------- (ii)	$1\frac{1}{2}$
	Adding (i) and (ii), we get	
	$2I = \int_0^{\pi}dx = x]_0^{\pi} = \pi,\ \therefore I = \frac{\pi}{2}$	$1\frac{1}{2}$
	Or	

	(b) $\int \frac{2x+1}{(x+1)^2(x-1)}dx = -\frac{3}{4}\int \frac{1}{x+1}\,dx + \frac{1}{2}\int \frac{1}{(x+1)^2}\,dx + \frac{3}{4}\int \frac{1}{x-1}\,dx$	$1\frac{1}{2}$
	$= -\frac{3}{4}\log\lvert x+1\rvert - \frac{1}{2(x+1)} + \frac{3}{4}\log\lvert x-1\rvert + C$	$1\frac{1}{2}$
	or, $= \frac{3}{4}\log\left\lvert\frac{x-1}{x+1}\right\rvert - \frac{1}{2(x+1)} + C$	
29.	(a) Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}$; $y(0) = 5$. **OR** (b) Solve the following differential equation : $x^2\,dy + y(x+y)\,dx = 0$	
Ans	**(a) Given differential equation is a linear order differential equation with:**	
	$P = -2x,\ Q = 3x^2e^{x^2}$	$\frac{1}{2}$
	Integrating Factor $= e^{\int -2xdx} = e^{-x^2}$	1
	The general solution is: $y\cdot e^{-x^2} = \int e^{-x^2}\cdot 3x^2e^{x^2}dx + C \Rightarrow y\cdot e^{-x^2} = x^3 + C$	1
	Putting $x = 0, y = 5$**, we get,** $C = 5$	
	$\therefore$ **The Particular solution is:** $y\cdot e^{-x^2} = x^3 + 5$ **or** $y = (x^3+5)e^{x^2}$	$\frac{1}{2}$
	Or	
	(b) $x^2dy + y(x+y)dx = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x} - \left(\frac{y}{x}\right)^2$	$\frac{1}{2}$
	Putting $\frac{y}{x} = v \Rightarrow y = vx,\ \frac{dy}{dx} = v + x\frac{dv}{dx}$	$\frac{1}{2}$
	$v + x\frac{dv}{dx} = -v - v^2,$	
	separating the variable and integrating	
	$\int \frac{1}{v^2+2v}dv = -\int \frac{1}{x}dx$	$\frac{1}{2}$
	$\Rightarrow \int \frac{1}{(v+1)^2-1}dv = -\int \frac{1}{x}dx$	$\frac{1}{2}$
	$\Rightarrow \frac{1}{2}\log\left\lvert\frac{v}{v+2}\right\rvert = \log\left\lvert\frac{C}{x}\right\rvert$	$\frac{1}{2}$

	The solution of the differential equation is, $\left\|\frac{y}{y+2x}\right\| = \frac{C^2}{x^2}$ **or** $x^2y = k(y+2x)$	$\frac{1}{2}$
30.	Find a vector of magnitude 4 units perpendicular to each of the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$ and hence verify your answer.	
Ans	**Let** $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} - \hat{k}$ **and** $\vec{c}$ **be the vector perpendicular to both** $\vec{a}$ **&** $\vec{b}$ **then,** $\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 3\hat{j} + 3\hat{k}$ **Let** $\vec{d}$ **the vector perpendicular to both the vectors** $\vec{a}$ **&** $\vec{b}$ **and having magnitude 4,** $\vec{d} = 4\hat{c} = 2\sqrt{2}\hat{j} + 2\sqrt{2}\hat{k}$ **(or** $-2\sqrt{2}\hat{j} - 2\sqrt{2}\hat{k}$ **)** **Verification:** $\left\|\vec{d}\right\| = \sqrt{8+8} = 4$, $\vec{d} \cdot \vec{a} = \vec{d} \cdot \vec{b} = 0 \Rightarrow \vec{d} \perp \vec{a}$ **and** $\vec{d} \perp \vec{b}$	**1** **1** **1**
31.	The random variable X has the following probability distribution where a and b are some constants : **X** | 1 | 2 | 3 | 4 | 5 **P(X)** | 0.2 | a | a | 0.2 | b If the mean E(X) = 3, then find values of a and b and hence determine $P(X \geq 3)$.	
Ans	$E(X) = 0.2 + 2a + 3a + 0.8 + 5b = 5a + 5b + 1$ $\sum p_i = 1 \Rightarrow 2a + b = 0.6$ $E(X) = 3 \Rightarrow 5a + 5b = 2$ **and,** **solving the two equations, we get,** $a = \frac{1}{5}, b = \frac{1}{5}$ $P(X \geq 3) = 1 - [P(X=1) + P(X=2)] = 1 - [0.2 + a] = 1 - \frac{2}{5} = \frac{3}{5}$	$\frac{1}{2}$ $\frac{1}{2}$ **1** **1**
	SECTION-D **(Question nos. 32 to 35 are Long Answer type questions carrying 5 marks each)**	
32.	(a) If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 0 & -3 \\ 1 & 2 & 0 \end{bmatrix}$, then find A^{-1} and hence solve the following system of equations : $x + 2y - 3z = 1$ $2x - 3z = 2$ $x + 2y = 3$ **OR**	

	(b) Find the product of the matrices $\begin{bmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{bmatrix}\begin{bmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{bmatrix}$ and hence solve the system of linear equations : $x + 2y - 3z = -4$ $2x + 3y + 2z = 2$ $3x - 3y - 4z = 11$	
Ans	**(a)** $\lvert A\rvert = 1(6)-2(3)-3(4) = -12 \neq 0$, $\therefore$ A^{-1} **exist**	1
	$\text{adj}A = \begin{bmatrix}6 & -6 & -6\\ -3 & 3 & -3\\ 4 & 0 & -4\end{bmatrix}$	2
	$\therefore A^{-1} = -\frac{1}{12}\begin{bmatrix}6 & -6 & -6\\ -3 & 3 & -3\\ 4 & 0 & -4\end{bmatrix}$	½
	The given system of equations can be written as $AX = B$, $X = \begin{bmatrix}x\\ y\\ z\end{bmatrix}, B = \begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$	
	$X = A^{-1}B \Rightarrow \begin{bmatrix}x\\ y\\ z\end{bmatrix} = -\frac{1}{12}\begin{bmatrix}6 & -6 & -6\\ -3 & 3 & -3\\ 4 & 0 & -4\end{bmatrix}\begin{bmatrix}1\\ 2\\ 3\end{bmatrix} = \begin{bmatrix}2\\ 1/2\\ 2/3\end{bmatrix}$	1½
	$\therefore$ The solution of the given system of equations is: $x = 2, y = \frac{1}{2}, z = \frac{2}{3}$	
	Or	
	(b) $\begin{bmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{bmatrix}\begin{bmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{bmatrix} = \begin{bmatrix}67 & 0 & 0\\ 0 & 67 & 0\\ 0 & 0 & 67\end{bmatrix}$	2½
	$\Rightarrow \begin{bmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{bmatrix}^{-1} = \frac{1}{67}\begin{bmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{bmatrix}$	1
	Solution of the system of equations is given by:	
	$\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{bmatrix}^{-1}\begin{bmatrix}-4\\ 2\\ 11\end{bmatrix} = \frac{1}{67}\begin{bmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{bmatrix}\begin{bmatrix}-4\\ 2\\ 11\end{bmatrix} = \begin{bmatrix}3\\ -2\\ 1\end{bmatrix}$,	1½
	$\therefore$ $x = 3, y = -2, z = 1$	

33.	Find the area of the region bounded by the curve $4x^2 + y^2 = 36$ using integration.	
Ans	**The given equation can be written as:** $\frac{x^2}{9}+\frac{y^2}{36}=1$**, which is an ellipse.**  **Correct Graph** **Area of the region bounded by the curve** $=4\times\frac{6}{3}\int_0^3\sqrt{9-x^2}\,dx$ $=8\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\frac{x}{3}\right]_0^3$ $=18\pi$	**1** **1½** **1½** **1**
34.	(a) Find the co-ordinates of the foot of the perpendicular drawn from the point (2, 3, –8) to the line $\frac{4-x}{2}=\frac{y}{6}=\frac{1-z}{3}$. Also, find the perpendicular distance of the given point from the line. **OR** (b) Find the shortest distance between the lines L_1 & L_2 given below : L_1 : The line passing through (2, –1, 1) and parallel to $\frac{x}{1}=\frac{y}{1}=\frac{z}{3}$ L_2 : $\vec{r}=\hat{i}+(2\mu+1)\hat{j}-(\mu+2)\hat{k}$.	
Ans	**(a) The standard form of the equation of the line is** $\frac{x-4}{-2}=\frac{y}{6}=\frac{z-1}{-3}$  **Let foot of the perpendicular from the point A$(2,3,-8)$ to the given line be B$(-2\lambda+4,6\lambda,-3\lambda+1)$** **D-ratios of AB is:** $-2\lambda+2,6\lambda-3,-3\lambda+9$ **As AB is perpendicular to the given line:** $-2(-2\lambda+2)+6(6\lambda-3)-3(-3\lambda+9)=0$ $\Rightarrow\lambda=1$ $\therefore$ **Foot of the perpendicular is: B$(2,6,-2)$** **Perpendicular distance = AB =** $3\sqrt{5}$ **Or**	**½** **½** **1** **1** **1** **1**

	(b) Equation of L_1 : $\vec{r} = 2\hat{i} - \hat{j} + \hat{k} + \lambda\left(\hat{i} + \hat{j} + 3\hat{k}\right)$ **Equation of L_2 :** $\vec{r} = \left(\hat{i} + \hat{j} - 2\hat{k}\right) + \mu\left(2\hat{j} - \hat{k}\right)$ **Taking** $\vec{a}_1 = 2\hat{i} - \hat{j} + \hat{k},\ \vec{b}_1 = \hat{i} + \hat{j} + 3\hat{k}$ $\vec{a}_2 = \hat{i} + \hat{j} - 2\hat{k},\ \vec{b}_2 = 2\hat{j} - \hat{k}$ $\vec{a}_2 - \vec{a}_1 = -\hat{i} + 2\hat{j} - 3\hat{k},\ \vec{b}_1 \times \vec{b}_2 = -7\hat{i} + \hat{j} + 2\hat{k}$ **Shortest Distance** $= \dfrac{\left\|\left(\vec{a}_2 - \vec{a}_1\right)\cdot\left(\vec{b}_1 \times \vec{b}_2\right)\right\|}{\left\|\vec{b}_1 \times \vec{b}_2\right\|} = \dfrac{1}{\sqrt{6}}$	**1** **½** **½** **½ + 1** **1½**
35.	Solve the following L.P.P. graphically : Maximise Z = 60*x* + 40y Subject to $x + 2y \le 12$ $2x + y \le 12$ $4x + 5y \ge 20$ $x, y \ge 0$	
Ans	 **Correct Graph**	**3½**
		1
	Max(Z) = 400 at x = 4, y = 4	**½**

Corner Points	Value of $Z = 60x + 40y$
$A(0,4)$	$Z = 160$
$B(0,6)$	$Z = 240$
$C(4,4)$	$Z = 400$
$D(6,0)$	$Z = 360$
$E(5,0)$	$Z = 300$

SECTION-E

(Question nos. 36 to 38 are source based/case based/passage based/integrated units of assessment questions carrying 4 marks each)

36. (a) Students of a school are taken to a railway museum to learn about railways heritage and its history.



An exhibit in the museum depicted many rail lines on the track near the railway station. Let L be the set of all rail lines on the railway track and R be the relation on L defined by

$R = \{(l_1, l_2) : l_1 \text{ is parallel to } l_2\}$

On the basis of the above information, answer the following questions :

(i) Find whether the relation R is symmetric or not.

(ii) Find whether the relation R is transitive or not.

(iii) If one of the rail lines on the railway track is represented by the equation $y = 3x + 2$, then find the set of rail lines in R related to it.

OR

(b) Let S be the relation defined by $S = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$ check whether the relation S is symmetric and transitive.

Ans

(a) (i) Let $(l_1, l_2) \in R \Rightarrow l_1 \parallel l_2 \Rightarrow l_2 \parallel l_1 \Rightarrow (l_2, l_1) \in R$, $\therefore$ R is a symmetric relation — 1

(ii) Let $(l_1, l_2), (l_2, l_3) \in R \Rightarrow l_1 \parallel l_2, l_2 \parallel l_3 \Rightarrow l_1 \parallel l_3 \Rightarrow (l_1, l_3) \in R$, $\therefore$ R is a transitive relation — 1

(iii) The set is $\{l : l \text{ is a line of type } y = 3x + c, c \in R\}$ — 2

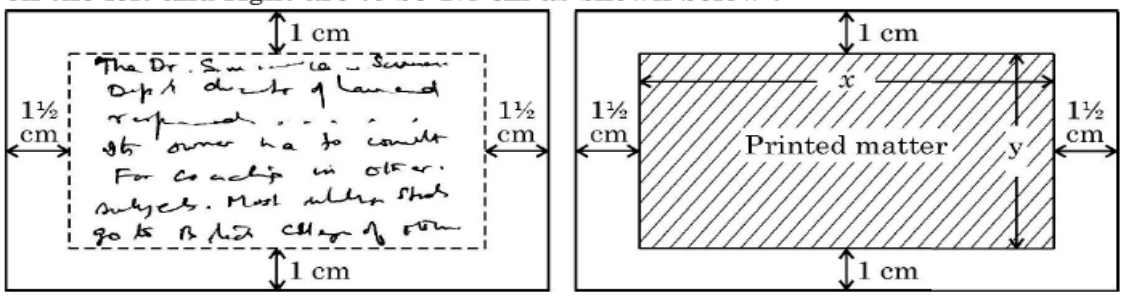
Or

(b) Let $(l_1, l_2) \in R \Rightarrow l_1 \perp l_2 \Rightarrow l_2 \perp l_1 \Rightarrow (l_2, l_1) \in R$, $\therefore$ R is a symmetric relation — 2

Let $(l_1, l_2), (l_2, l_3) \in R \Rightarrow l_1 \perp l_2, l_2 \perp l_3 \Rightarrow l_1 \parallel l_3 \Rightarrow (l_1, l_3) \notin R$, $\therefore$ R is not a transitive relation — 2

** Due to printing error Part (a) or Part(b), both parts be taken as independent questions of 4 marks each

37. A rectangular visiting card is to contain 24 sq.cm. of printed matter. The margins at the top and bottom of the card are to be 1 cm and the margins on the left and right are to be 1½ cm as shown below :



On the basis of the above information, answer the following questions :

(i) Write the expression for the area of the visiting card in terms of x.

(ii) Obtain the dimensions of the card of minimum area.

Ans

(i) Let $A(x)$ be the area of the visiting card then,

$$\text{As } xy = 24,\ A(x) = (x+3)(y+2) = 2x + 3y + xy + 6 = 2x + \frac{72}{x} + 30$$ **2**

(ii) $A'(x) = 2 - \frac{72}{x^2}$ **and** $A''(x) = \frac{144}{x^3}$, **1**

solving $A'(x) = 0 \Rightarrow x = 6$ **is the critical point.** ½

$A''(6) = \frac{144}{6} > 0$, **$\therefore$ Area of the card is minimum at** $x = 6, y = 4$

The dimension of the card with minimum area is Length = 9 cm, Breadth = 6 cm ½

38. A departmental store sends bills to charge its customers once a month. Past experience shows that 70% of its customers pay their first month bill in time. The store also found that the customer who pays the bill in time has the probability of 0.8 of paying in time next month and the customer who doesn't pay in time has the probability of 0.4 of paying in time the next month.

Based on the above information, answer the following questions :

(i) Let E_1 and E_2 respectively denote the event of customer paying or not paying the first month bill in time.
Find $P(E_1)$, $P(E_2)$.

(ii) Let A denotes the event of customer paying second month's bill in time, then find $P(A|E_1)$ and $P(A|E_2)$.

(iii) Find the probability of customer paying second month's bill in time.

OR

(iii) Find the probability of customer paying first month's bill in time if it is found that customer has paid the second month's bill in time.

Ans	**(i)** $P(E_1) = \frac{7}{10} = 0.7,\ P(E_2) = \frac{3}{10} = 0.3$	**1**
	(ii) $P(A \mid E_1) = 0.8, P(A \mid E_2) = 0.4$	**1**
	(iii) $P(A) = P(E_1) \cdot P(A \mid E_1) + P(E_2) \cdot P(A \mid E_2) = 0.7 \times 0.8 + 0.3 \times 0.4 = 0.68$ **or** $\frac{17}{25}$	**2**
	Or	
	(iii) $P(A) = \frac{P(E_1) \cdot P(A \mid E_1)}{P(E_1) \cdot P(A \mid E_1) + P(E_2) \cdot P(A \mid E_2)} = \frac{14}{17}$	**2**

MARKING SCHEME
MATHEMATICS (Subject Code–041)
(PAPER CODE: 65(B))

Q. No.	EXPECTED OUTCOMES/VALUE POINTS	Marks
	SECTION A **Questions no. 1 to 18 are multiple choice questions (MCQs) and questions number 19 and 20 are Assertion-Reason based questions of 1 mark each**	
1.	If $A = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$, then the value of x, for which A is an identity matrix, is (A) $\frac{\pi}{2}$ (B) π (C) 0 (D) $\frac{3\pi}{2}$	
Sol.	(C) 0	**1**
2.	If the matrix $A = \begin{bmatrix} 0 & 5 & -7 \\ a & 0 & 3 \\ b & -3 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the values of 'a' and 'b' are : (A) $a = 5, b = 3$ (B) $a = 5, b = -7$ (C) $a = -5, b = -7$ (D) $a = -5, b = 7$	
Sol.	(D) $a = -5, b = 7$	**1**
3.	If $\begin{vmatrix} x+2 & x-4 \\ x-2 & x+3 \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 1 & 3 \end{vmatrix}$, then the value of x is : (A) 1 (B) 2 (C) -2 (D) -1	

Sol.	(B) 2	**1**
4.	If $\begin{bmatrix} 8 & 14 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} X$, then matrix X is : (A) $\begin{bmatrix} 3 & 7 \\ 2 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 0 \\ 7 & 3 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix}$ (D) $\begin{bmatrix} 2 & 0 \\ -3 & 7 \end{bmatrix}$	
Sol.	(C) $\begin{bmatrix} 2 & 0 \\ 3 & 7 \end{bmatrix}$	**1**
5.	The value of k, for which $f(x) = \begin{cases} \dfrac{\sqrt{3}\cos x + \sin x}{3x + \dfrac{\pi}{2}}, & x \neq -\dfrac{\pi}{3} \\ k, & x = -\dfrac{\pi}{3} \end{cases}$ is continuous at $x = -\dfrac{\pi}{3}$, is : (A) $\dfrac{2}{3}$ (B) $-\dfrac{2}{3}$ (C) $\dfrac{3}{2}$ (D) 6	
Sol.	Since correct answer is not in the options given So, 1 mark may be given to all who attempted this question	**1**
6.	Let the vectors $\vec{a}$ and $\vec{b}$ be such that $\lvert \vec{a} \rvert = \sqrt{3}$ and $\lvert \vec{b} \rvert = \dfrac{2}{\sqrt{3}}$, then $\vec{a} \times \vec{b}$ is a unit vector, if the angle between $\vec{a}$ and $\vec{b}$ is : (A) $\dfrac{\pi}{3}$ (B) $\dfrac{\pi}{4}$ (C) $\dfrac{\pi}{6}$ (D) $\dfrac{\pi}{2}$	
Sol.	(C) $\frac{\pi}{6}$	**1**

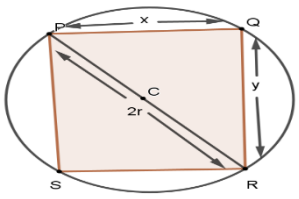
7.	If $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$, then the projection of $(\vec{c} - \vec{b})$ along $\vec{a}$ is : (A) 15 (B) 5 (C) $\frac{2}{3}$ (D) 1	
Sol.	(B) 5	**1**
8.	The angle between the lines $\frac{x+1}{2} = \frac{2-y}{-5} = \frac{z}{4}$ and $\frac{x-3}{1} = \frac{y-7}{2} = \frac{5-z}{3}$ is : (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$	
Sol.	(B) $\frac{\pi}{2}$	**1**
9.	The Cartesian equations of a line are given as $6x - 2 = 3y + 1 = 2z - 2$ The direction ratios of the line are : (A) 2, – 1, 3 (B) 1, – 2, – 3 (C) 1, 2, 3 (D) 3, 1, 2	
Sol.	(C) 1, 2, 3	**1**
10.	The solution set of the inequation $2x + 3y < 6$ is : (A) open half-plane not containing origin (B) whole xy-plane except the points lying on the line $2x + 3y = 6$ (C) open half-plane containing origin (D) half-plane containing the origin and the points lying on the line $2x + 3y = 6$	
Sol.	(C) open half-plane containing origin	**1**

11.	The maximum value of the objective function $z = 3x + 5y$ subject to the constraints $x \geq 0$, $y \geq 0$ and $4x + 3y \leq 12$ is : (A) 15 (B) 29 (C) 9 (D) 20	
Sol.	(D) 20	**1**
12.	If the points A(3, – 2), B(k, 2) and C(8, 8) are collinear, then the value of k is : (A) 2 (B) – 3 (C) 5 (D) – 4	
Sol.	(C) 5	**1**
13.	If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then $(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$ is equal to : (A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$	
Sol.	(D) $-\frac{3}{2}$	**1**
14.	$\int \frac{\cos 2x}{\sin^2 x \cos^2 x}\, dx$ is equal to : (A) $\cot x + \tan x + c$ (B) $-\cot x + \tan x + c$ (C) $\cot x - \tan x + c$ (D) $-\cot x - \tan x + c$	
Sol.	(D) $-\cot x - \tan x + c$	**1**

15.	The solution of the differential equation $\frac{dy}{dx} = 1 - x + y - xy$ is : (A) $\log \lvert 1 + y \rvert = x - \frac{x^2}{2} + c$ (B) $\log \lvert 1 + y \rvert = -x + \frac{x^2}{2} + c$ (C) $e^y = x - \frac{x^2}{2} + c$ (D) $e^{(1+y)} = -x + \frac{x^2}{2} + c$	
Sol.	(A) $\log\lvert 1 + y\rvert = x - \frac{x^2}{2} + c$	**1**
16.	The degree of the differential equation $x\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + y^5 = 0$ is : (A) 2 (B) 3 (C) 4 (D) 5	
Sol.	(B) 3	**1**
17.	The integrating factor of the differential equation $\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$ is : (A) $e^{\sec x}$ (B) $\sec x + \tan x$ (C) $\sec x$ (D) $\cos x$	
Sol.	(C) $\sec x$	**1**
18.	The probabilities of A, B and C solving a problem are $\frac{1}{3}$, $\frac{1}{5}$ and $\frac{1}{6}$ respectively. The probability that the problem is solved, is : (A) $\frac{4}{9}$ (B) $\frac{5}{9}$ (C) $\frac{1}{90}$ (D) $\frac{1}{3}$	

Sol.	(B) $\frac{5}{9}$	**1**
	Questions number ***19*** *and* ***20*** *are Assertion and Reason based questions. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.* (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	
19.	*Assertion (A) :* $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$ *Reason (R) :* $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$	
Sol.	(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).	**1**
20.	*Assertion (A) :* If the side of a square is increasing at the rate of 0·2 cm/s, then the rate of increase of its perimeter is 0·8 cm/s. *Reason (R) :* Perimeter of a square = 4 (side).	
Sol.	(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).	**1**
	SECTION B **This section comprises of Very Short Answer (VSA) type questions of 2 marks each.**	
21(a).	Find the value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.	

Sol.	$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ $= \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6}$ $= \frac{3\pi}{4}$	**1½** **½**
	OR	
21(b).	Find the domain of the function $y = \cos^{-1}(x^2 - 4)$.	
Sol.	$-1 \le x^2 - 4 \le 1$ $3 \le x^2 \le 5$ $\Rightarrow$ domain $= [-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$	**1** **1**
22(a).	Differentiate $\cot^{-1}(\sqrt{1+x^2} + x)$ w.r.t. x.	
Sol.	$y = \cot^{-1}(\sqrt{1+x^2} + x)$, (Put $x = \cot\theta$) $= \cot^{-1}(\operatorname{cosec}\theta + \cot\theta) = \cot^{-1}\left(\frac{1+\cos\theta}{\sin\theta}\right)$ $= \cot^{-1}(\cot\frac{\theta}{2}) = \frac{1}{2}\cot^{-1}x$ $\frac{dy}{dx} = -\frac{1}{2(1+x^2)}$	½ ½ ½ ½
22(b).	If $(\cos x)^y = (\cos y)^x$, find $\frac{dy}{dx}$.	
Sol.	$(\cos x)^y = (\cos y)^x$ $\Rightarrow y \log\cos x = x \log\cos y$ Differentiating, $\frac{dy}{dx}\log\cos x + y(-\tan x) = \log\cos y + x(-\tan y)\frac{dy}{dx}$	½ **1**

	$\frac{dy}{dx} = \frac{y\tan x + \log\cos y}{x\tan y + \log\cos x}$	½
23.	Find the intervals on which the function $f(x) = 10 - 6x - 2x^2$ is (a) strictly increasing (b) strictly decreasing.	
Sol.	$f'(x) = -6 - 4x$ $f'(x) = 0$ gives $x = -\frac{3}{2}$ Intervals are $(-\infty, -\frac{3}{2})$ and $(-\frac{3}{2}, \infty)$ $f'(x) > 0$ for $x < -\frac{3}{2}$, f is strictly increasing in $(-\infty, -\frac{3}{2})$ *or* $(-\infty, -\frac{3}{2}]$ $f'(x) < 0$ for $x > -\frac{3}{2}$, f is strictly decreasing in $(-\frac{3}{2}, \infty)$ *or* $[-\frac{3}{2}, \infty)$	½ ½ ½ ½
24.	Show that of all rectangles inscribed in a given circle, the square has the maximum area.	
Sol.	Let the rectangle with sides x and y has maximum area $\therefore A = x.y$ where $x^2 + y^2 = 4r^2, r$ being the radius $\therefore A = x\sqrt{4r^2 - x^2}$ or $Z = x^2(4r^2 - x^2) = 4r^2x^2 - x^4$, where $Z = A^2$ $\frac{dZ}{dx} = 8r^2x - 4x^3$, $\frac{dZ}{dx} = 0 \implies x = \sqrt{2}r$ $\therefore y = \sqrt{4r^2 - 2r^2} = \sqrt{2}r$ $\frac{d^2Z}{dx^2} = 8r^2 - 12x^2 < 0$ for $x^2 = 2r$	 ½ ½ ½

	$\therefore x = y = \sqrt{2}r$ Therefore, of all the rectangles inscribed in a given circle, the square has the maximum area.	½
25.	Find : $\int \text{cosec}^3 (3x+1) \cot (3x+1)\ dx$	
Sol.	$I = \int \text{cosec}^3(3x+1)\cot(3x+1)\,dx$ Put $\text{cosec}\,(3x+1) = t$, $-\text{cosec}\,(3x+1)\cot(3x+1)\,dx = \frac{1}{3}dt$ $I = -\frac{1}{3}\int t^2 dt = -\frac{1}{9}t^3 + c$ $= -\frac{1}{9}\text{cosec}^3(3x+1) + c$	**1** ½ ½
	SECTION C **This section comprises of Short Answer (SA) type questions of 3 marks each.**	
26.	If $x = a\cos\theta$ and $y = b\sin\theta$, then prove that $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2y^3}$.	
Sol.	$\frac{dx}{d\theta} = -a\sin\theta, \frac{dy}{d\theta} = b\cos\theta \implies \frac{dy}{dx} = -\frac{b}{a}\cot\theta$ $\frac{d^2y}{dx^2} = -\frac{b}{a}(-\text{cosec}^2\theta).\frac{d\theta}{dx} = \frac{b}{a}\text{cosec}^2\theta\left(-\frac{1}{a\sin\theta}\right) = \frac{-b}{a^2\sin^3\theta}$ $= -\frac{b^4}{a^2y^3}$	**1½** **1** ½
27.	Find : $\int \frac{2x}{(x^2+1)(x^2+3)}\,dx$	

Sol.	$I = \int \frac{2x}{(x^2+1)(x^2+3)}\,dx$, (Put $x^2 = t, 2xdx = dt$)	½
	$= \int \left(\frac{\frac{1}{2}}{t+1} - \frac{\frac{1}{2}}{t+3} \right) dt$	**1**
	$= \frac{1}{2}[\log(t+1) - \log(t+3)] + c$	**1**
	$= \frac{1}{2}[\log(x^2+1) - \log(x^2+3)] + c$ or $\frac{1}{2}\log\left\|\frac{x^2+1}{x^2+3}\right\| + c$	½
28(a).	Evaluate : $\int_{-6}^{6} \|x+2\|\,dx$	
Sol.	$I = \int_{-6}^{6} \|x+2\|\,dx = \int_{-6}^{-2} \|x+2\|\,dx + \int_{-2}^{6} \|x+2\|\,dx$	**1**
	$= \int_{-6}^{-2} -(x+2)\,dx + \int_{-2}^{6} (x+2)\,dx$	½
	$= -\left[\frac{(x+2)^2}{2}\right]_{-6}^{-2} + \left[\frac{(x+2)^2}{2}\right]_{-2}^{6}$	**1**
	$= -(0-8) + (32) = 40$	½
28(b).	Find : $\int \left(\frac{4-x}{x^5}\right) e^x\,dx$	
Sol.	$I = \int \left(\frac{4-x}{x^5}\right) e^x dx$	
	$= \int (4x^{-5} - x^{-4}) e^x\,dx$ (if $f(x) = -x^{-4}$, then $f'(x) = 4x^{-5}$)	**1½**
	$I = -x^{-4}e^x + c$ or $-\frac{e^x}{x^4} + c$	**1½**

29(a).	Find the particular solution of the differential equation $2xy\,\frac{dy}{dx} = x^2 + 3y^2$, given that $y(1) = 0$.	
Sol.	$\frac{dy}{dx} = \frac{x^2+3y^2}{2xy} = \frac{1 + 3\left(\frac{y}{x}\right)^2}{2\frac{y}{x}}$ Put $\frac{y}{x} = v \Rightarrow y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ $v + x\frac{dv}{dx} = \frac{1+3v^2}{2v} \Rightarrow x\frac{dv}{dx} = \frac{1+v^2}{2v}$ $\Rightarrow \int \frac{2v}{1+v^2}dv = \int \frac{dx}{x} \Rightarrow \log(1+v^2) = \log x + \log c$ $\Rightarrow 1 + \left(\frac{y}{x}\right)^2 = cx, x = 1, y = 0 \Rightarrow c = 1$ $\Rightarrow x^2 + y^2 = x^3$ is the required solution	½ **1** **1** ½
29(b).	Solve the differential equation $\frac{dy}{dx} + 2y\tan x = \sin x$, given that $y = 0$, when $x = \frac{\pi}{3}$.	
Sol.	$\frac{dy}{dx} + 2y\tan x = \sin x$ I.F. $= e^{\int 2\tan x dx} = e^{2\log\sec x} = \sec^2 x$ $\therefore y\sec^2 x = \int \sin x \sec^2 x\,dx = \int \sec x \tan x\,dx$ $= \sec x + c$ $y = 0, x = \frac{\pi}{3} \Rightarrow 0 = 2 + c \Rightarrow c = -2$ $\Rightarrow y\sec^2 x = \sec x - 2$ Or $y = \cos x - 2\cos^2 x$	**1** **1½** ½

30.	The corner points of the feasible region determined by the system of linear constraints are A(0, 40), B(20, 40), C(60, 20) and D(60, 0). The objective function of the L.P.P. is z = 4x + 3y. Find the point of the feasible region at which the value of objective function is maximum and the point at which the value is minimum. Hence, find the maximum and the minimum values.	
Sol.	$Z = 4x + 3y$ Z_A 120 Z_B 200 Z_C 300 Z_D 240 Max Z = 300 at point C Min Z = 120 at point A	**2** **1**
31(a).	A card is randomly drawn from a well-shuffled pack of 52 playing cards. Events A and B are defined as under : A : Getting a card of diamond B : Getting a queen Determine whether the events A and B are independent or not.	
Sol.	$P(A) = P(\text{diamond}) = \frac{13}{52} = \frac{1}{4}$ $P(B) = P(\text{queen}) = \frac{4}{52} = \frac{1}{13}$ $P(A \cap B) = P(\text{queen of diamond}) = \frac{1}{52}$ Since $P(A).P(B) = \frac{1}{52} = P(A \cap B)$ $\Rightarrow$ A and B are independent events.	½ ½ ½ **1** ½
	OR	

31(b). Find the probability distribution of the number of doublets in three throws of a pair of dice.

Sol. $p = P(\text{doublet}) = \frac{1}{6}, \quad q = 1 - \frac{1}{6} = \frac{5}{6}$ **1**

Let X denote the number of doublets. X can take values 0, 1, 2 or 3 **1**

X	0	1	2	3
P(X)	$\left(\frac{5}{6}\right)^3 = \frac{125}{216}$	$3 \times \frac{1}{6} \times \left(\frac{5}{6}\right)^2 = \frac{75}{216}$	$3 \times \left(\frac{1}{6}\right)^2 \times \frac{5}{6} = \frac{15}{216}$	$\left(\frac{1}{6}\right)^3 = \frac{1}{216}$

1

SECTION D

This section comprises of Long Answer (LA) type questions of 5 marks each.

32(a). Let $A = \{x \mid x \in Z, 0 \le x \le 12\}$. Show that the relation $R = \{(a, b) : a, b \in A, (a - b) \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of elements related to 2.

$R = \{(a, b) : a, b \in A, (a - b) \text{ is divisible by } 4\}$

(i) For every $a \in A$ we have $a - a = 0$ which is divisible by 4 **1**

$\Rightarrow (a, a) \in R$ for all $a \in A$ Hence R is reflexive

(ii) Let $(a, b) \in R, a, b \in A$ **1**

$\Rightarrow (a - b)$ is divisible by 4 $\Rightarrow (b - a)$ is also divisible by 4

$\therefore (b, a) \in R$ Hence R is symmetric

(iii) Let $(a, b) \in R$ and $(b, c) \in R, : a, b, c \in A$ **2**

Then $(a - b)$ and $(b - c)$ are divisible by 4

$\Rightarrow a - c = (a - b) + (b - c)$ which is divisible by 4

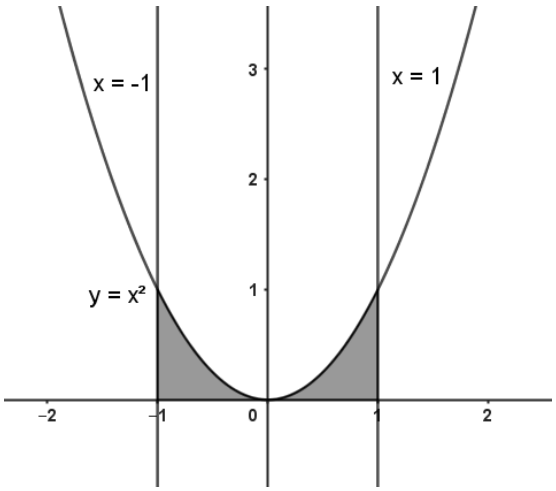
$\therefore (a, c) \in R$ Hence R is transitive

R is an equivalence relation

$[2] = \{2, 6, 10\}$ **1**

OR

32(b).	Let $A = \mathbb{R} - \{4\}$ and $B = \mathbb{R} - \{1\}$ and let function $f : A \to B$ be defined as $f(x) = \dfrac{x-3}{x-4}$ for $\forall\, x \in A$. Show that f is one-one and onto.	
Sol.	$f(x) = \frac{x-3}{x-4}$ Let $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$ $\therefore \frac{x_1-3}{x_1-4} = \frac{x_2-3}{x_2-4}$ $\Longrightarrow x_1 = x_2$ $\Longrightarrow$ f is one-one Let $y = \frac{x-3}{x-4} \in \mathbb{R} - \{4\}$ Then $xy - 4y = x - 3$ $\Longrightarrow x = \frac{4y-3}{y-1}$ Range f = B = Codomain f $\Longrightarrow$ f is onto	**2½** **2½**
33.	Using matrices, solve the following system of linear equations : $3x + 4y + 2z = 8\ ;\ 2y - 3z = 3\ ;\ x - 2y + 6z = -2$	
Sol.	$\begin{bmatrix}3 & 4 & 2\\ 0 & 2 & -3\\ 1 & -2 & 6\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}8\\ 3\\ -2\end{bmatrix}$ or $AX = B$ $\lvert A\rvert = 2 \neq 0$ $\text{adj}(A) = \begin{bmatrix}6 & -28 & -16\\ -3 & 16 & 9\\ -2 & 10 & 6\end{bmatrix}$ $\Longrightarrow A^{-1} = \frac{1}{2}\begin{bmatrix}6 & -28 & -16\\ -3 & 16 & 9\\ -2 & 10 & 6\end{bmatrix}$ $\therefore \begin{bmatrix}x\\ y\\ z\end{bmatrix} = \frac{1}{2}\begin{bmatrix}6 & -28 & -16\\ -3 & 16 & 9\\ -2 & 10 & 6\end{bmatrix}\begin{bmatrix}8\\ 3\\ -2\end{bmatrix} = \begin{bmatrix}-2\\ 3\\ 1\end{bmatrix}$ $\Longrightarrow x = -2, y = 3, z = 1$	½ **1** **2** **1** ½

34.	Using integration, find the area of the region bounded by the curve $y = x^2$, $x = -1$, $x = 1$ and the x-axis.	
Sol.	$y = x^2$, $x = 1, x = 0$ and $x = -1$  Required area $= 2\int_0^1 x^2 dx$ $= 2\left[\frac{x^3}{3}\right]_0^1$ $= 2\left(\frac{1}{3}\right) = \frac{2}{3}$	**2** **2** **1**
35(a).	Write the vector equations of the following lines and hence find the shortest distance between them : $\frac{x+1}{2} = \frac{y+1}{-6} = \frac{z+1}{1}$ and $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$	
Sol.	Vector equations are $\vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \lambda(2\hat{i} - 6\hat{j} + \hat{k})$ and $\vec{r} = (3\hat{i} + 5\hat{j} + 7\hat{k}) + \mu(\hat{i} - 2\hat{j} + \hat{k})$ $\overrightarrow{a_2} - \overrightarrow{a_1} = (4\hat{i} + 6\hat{j} + 8\hat{k})$	**1** **1** **½**

	$\overrightarrow{b_1} \times \overrightarrow{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix} = -4\hat{i} - \hat{j} + 2\hat{k}$ SD $= \frac{\lvert(\overrightarrow{a_2} - \overrightarrow{a_1}).(\overrightarrow{b_1} \times \overrightarrow{b_2})\rvert}{\lvert\overrightarrow{b_1} \times \overrightarrow{b_2}\rvert}$ $= \frac{\lvert -16-6+16\rvert}{\sqrt{16+1+4}} = \frac{6}{\sqrt{21}}$	**1½** **1**
	OR	
35(b).	Find the length and the coordinates of the foot of the perpendicular drawn from the point P(5, 9, 3) to the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Also, find the coordinates of the image of the point P in the given line.	
Sol.	Any point on the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} = \lambda$ is $Q(2\lambda + 1, 3\lambda + 2, 4\lambda + 3)$ DRs of $\overrightarrow{PQ}$ are $(2\lambda - 4, 3\lambda - 7, 4\lambda)$ $\overrightarrow{PQ} \perp$ given line $\Rightarrow 2(2\lambda - 4) + 3(3\lambda - 7) + 4(4\lambda) = 0$ $\Rightarrow \lambda = 1$ $\therefore$ Point Q is (3,5,7) $PQ = \sqrt{4 + 16 + 16} = 6$ coordinates of image P' are (1,1,11)	**1** **1** **1** **1** **½** **½**
	SECTION E **This section comprises of 3 case-study based questions of 4 marks each.**	
36.	**Case Study – 1** The relation between the height of the plant (y in cm) with respect to exposure to sunlight is governed by the relation $y = 4x - \frac{1}{2}x^2$, where x is the number of days it is exposed to sunlight. Based on the above, answer the following questions :	

	(i) Find the rate of growth of the plant with respect to sunlight. *1* (ii) What is the number of days it will take for the plant to grow to the maximum height ? *2* (iii) What is the maximum height of the plant ? *1*	
Sol.	(i) $y = 4x - \frac{1}{2}x^2 \Longrightarrow \frac{dy}{dx} = (4 - x)$ cm/day (ii) For maximum height, $\frac{dy}{dx} = 0 \Longrightarrow x = 4$ days and $\frac{d^2y}{dx^2} < 0$ (iii) Maximum height $= y(4) = 16 - \frac{1}{2}(16) = 8$ cm	**1** **1½ +½** **1**
37.	**Case Study – 2** A cricket match is organised between two clubs P and Q for which a team from each club is chosen. Remaining players of club P and club Q are respectively sitting along the lines AB and CD, where the points are A(3, 4, 0), B(5, 3, 3), C(6, – 4, 1) and D(13, – 5, – 4). Based on the above, answer the following questions : (i) Write the direction ratios of vector $\overrightarrow{AB}$. *1* (ii) Write a unit vector in the direction of $\overrightarrow{CD}$. *1* (iii) (a) Find the angle between vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$. *2* **OR** (iii) (b) Write a vector perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{CD}$. *2*	

Sol.	(i) $\overrightarrow{AB} = \vec{b} - \vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$ $\therefore$ DRs of vector $\overrightarrow{AB}$ are $2, -1, 3$ (ii) $\overrightarrow{CD} = 7\hat{i} - \hat{j} - 5\hat{k}$, $\widehat{CD} = \frac{7}{5\sqrt{3}}\hat{i} - \frac{1}{5\sqrt{3}}\hat{j} - \frac{5}{5\sqrt{3}}\hat{k}$ (iii) (a) $\overrightarrow{AB}.\overrightarrow{CD} = 14 + 1 - 15 = 0$ $\overrightarrow{AB} \perp \overrightarrow{CD} \Rightarrow \theta = 90°$ **OR** (iii) (b) $\overrightarrow{AB} \times \overrightarrow{CD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 3 \\ 7 & -1 & -5 \end{vmatrix} = 8\hat{i} + 31\hat{j} + 5\hat{k}$	**1** ½ ½ **1** ½ ½ **1** **1**
38.	**Case Study – 3** A coach is training 3 players. He observes that player A can hit a target 4 times in 5 shots, player B can hit 3 times in 4 shots and player C can hit 2 times in 3 shots. Based on the above, answer the following questions : (i) Find the probability that all three players miss the target. *1* (ii) Find the probability that all of them hit the target. *1* (iii) (a) Find the probability that only one of them hits the target. *2* **OR** (iii) (b) Find the probability that exactly two of them hit the target. *2*	
Sol.	(i) P(all will miss the target) $= \left(\frac{1}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{3}\right) = \frac{1}{60}$ (ii)P(all hit the target) $= \left(\frac{4}{5}\right)\left(\frac{3}{4}\right)\left(\frac{2}{3}\right) = \frac{2}{5}$ (iii)(a) P(only one hit the target) $= \left(\frac{4}{5}\right)\left(\frac{1}{4}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{5}\right)\left(\frac{3}{4}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{5}\right)\left(\frac{1}{4}\right)\left(\frac{2}{3}\right)$ $= \frac{4}{60} + \frac{3}{60} + \frac{2}{60} = \frac{9}{60} = \frac{3}{20}$ **OR**	**1** **1** **1** ½ ½
	(iii)(b) P(Exactly two hits) $= \left(\frac{4}{5}\right)\left(\frac{3}{4}\right)\left(\frac{1}{3}\right) + \left(\frac{1}{5}\right)\left(\frac{3}{4}\right)\left(\frac{2}{3}\right) + \left(\frac{4}{5}\right)\left(\frac{1}{4}\right)\left(\frac{2}{3}\right)$ $= \frac{12}{60} + \frac{6}{60} + \frac{8}{60} = \frac{26}{60} = \frac{13}{30}$	**1** ½ ½